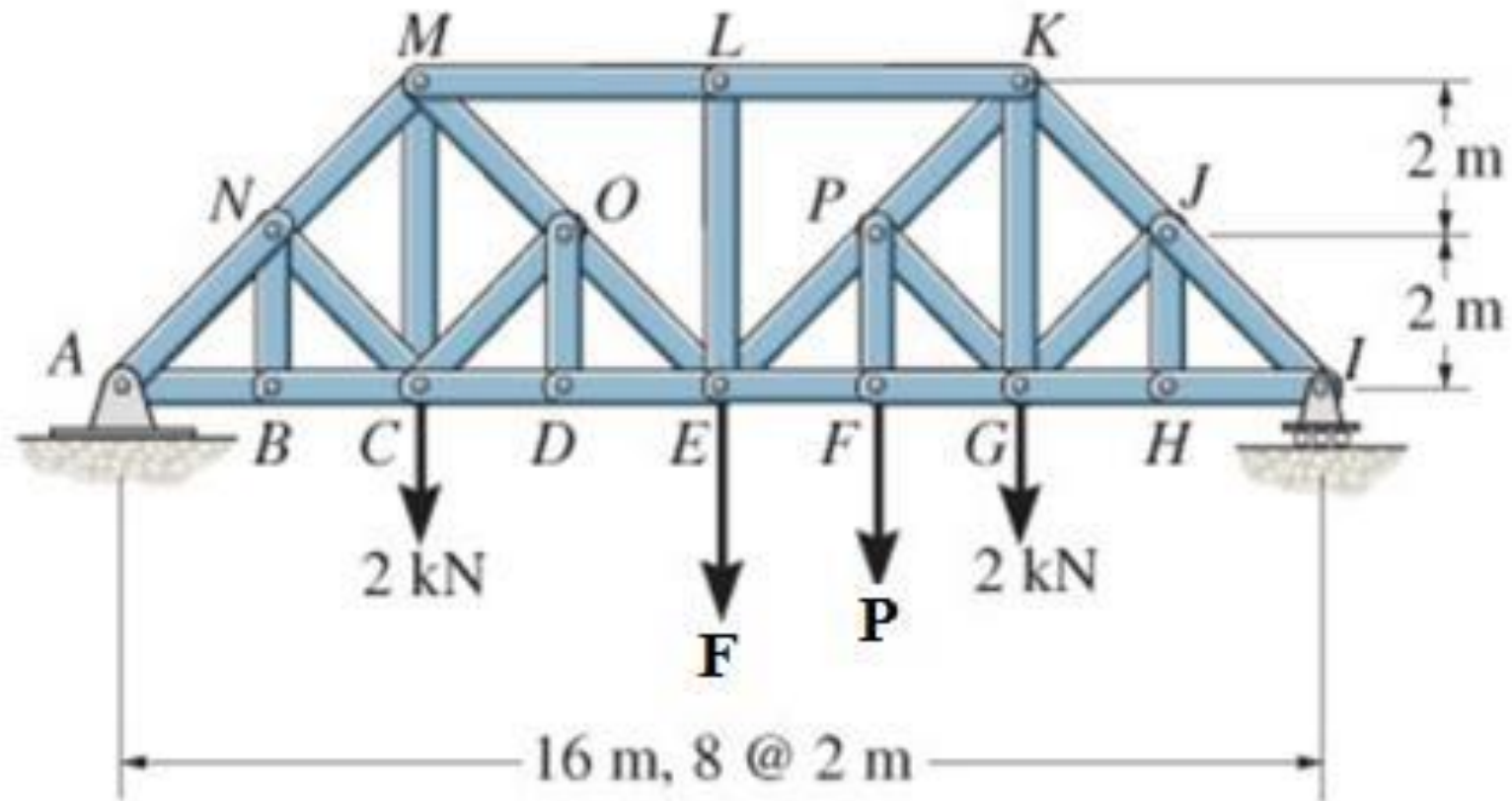


Method of Members



Learning Objectives

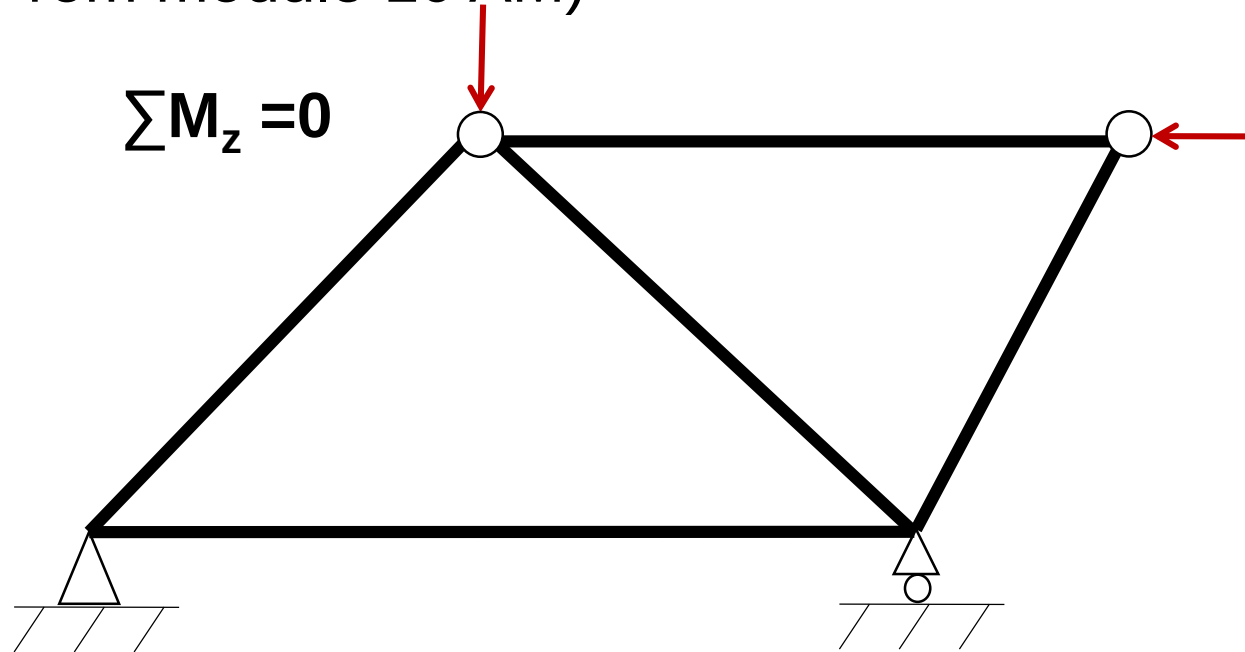
Students will be able to:

1. Identify the external forces VS internal body forces
2. Identify a two-force member
3. Understand the relationship between the forces acting on a two member
4. Draw the Free Body Diagrams (FBDs) for the system and the individual members in the system
5. Apply static equilibrium analysis to solve for all unknowns

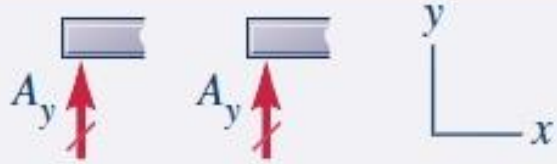
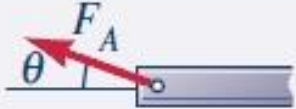
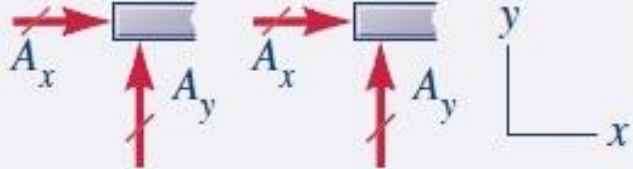
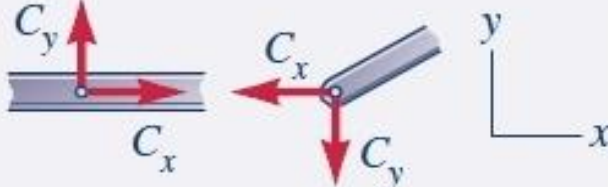
- Truss problems

- Deals with 2-force members
- i.e. forces acting at only 2 points in a member with no couples (moment)
- Method of joints (From module 10 AM)

- $\sum F_x = 0;$ $\sum F_y = 0,$ $\sum M_z = 0$

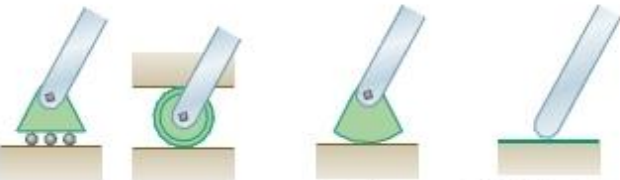
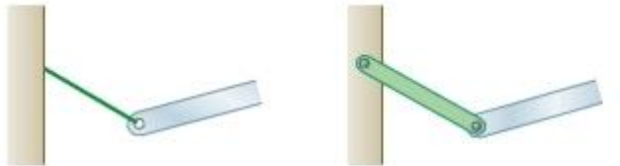
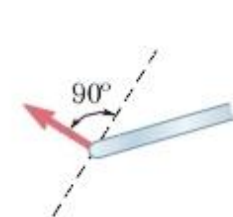


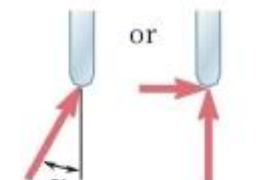
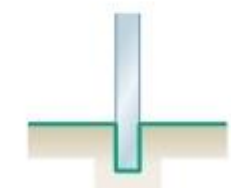
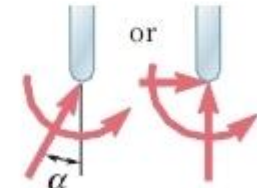
External forces

1. Roller support		
2. Cable or rod		
3. Pin support		
4. Cantilever support (fixed end)		
7. Pinned connection		

Craig, R.R.(2011). Mechanics of Material (Third Edition). John Wiley & Sons.

External forces

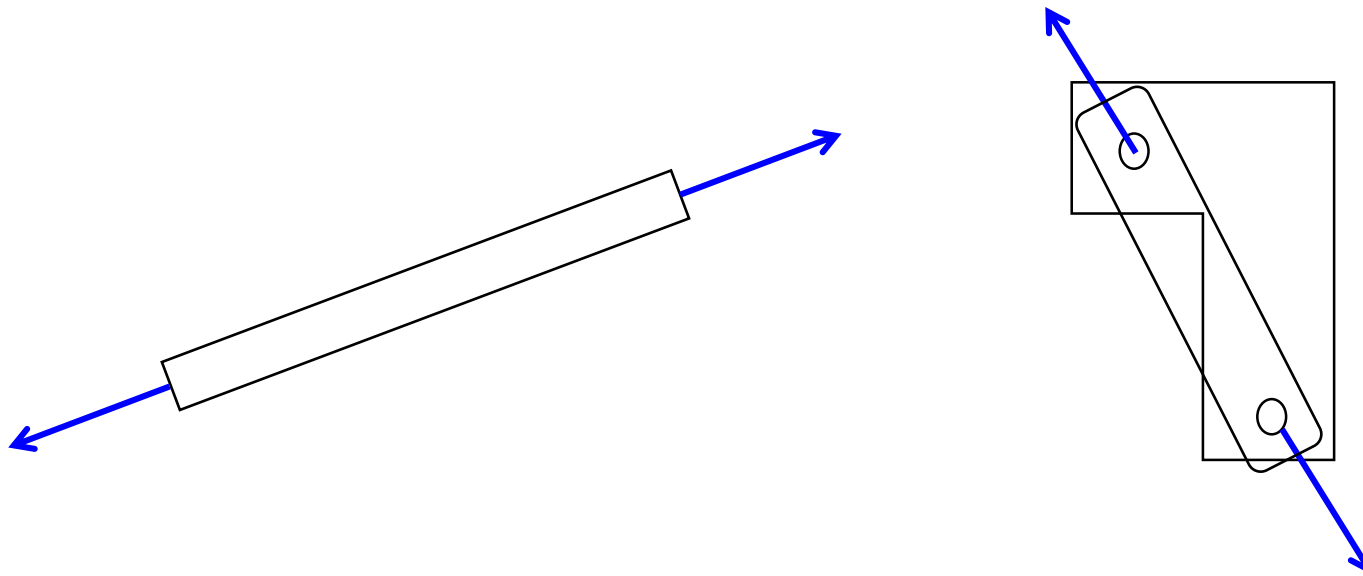
Support or Connection	Reaction	Number of Unknowns
 Rollers Rocker Frictionless surface	 Force with known line of action	1
 Short cable Short link	 Force with known line of action	1
 Collar on frictionless rod Frictionless pin in slot	 Force with known line of action	1

Support or Connection	Reaction	Number of Unknowns
 Frictionless pin or hinge Rough surface	 Force of unknown direction	2
 Fixed support	 Force and couple	3

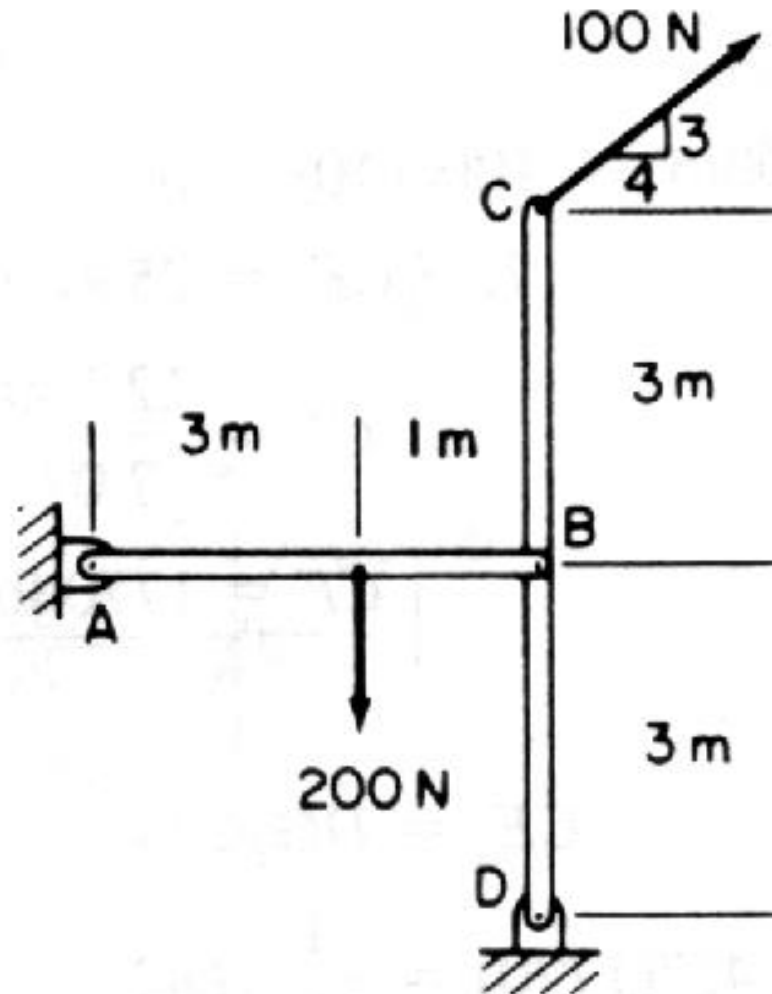
Beer, F.P. et al. (2011). *Statics and Mechanics of Material*. New York: McGraw Hill.

Two-force member

- A 2-force member has collinear force and recognising this greatly simplifies the problem solving process.
- 2 force member can only have tension or compression forces.
- Note: Only tension or compression can take place at one time.



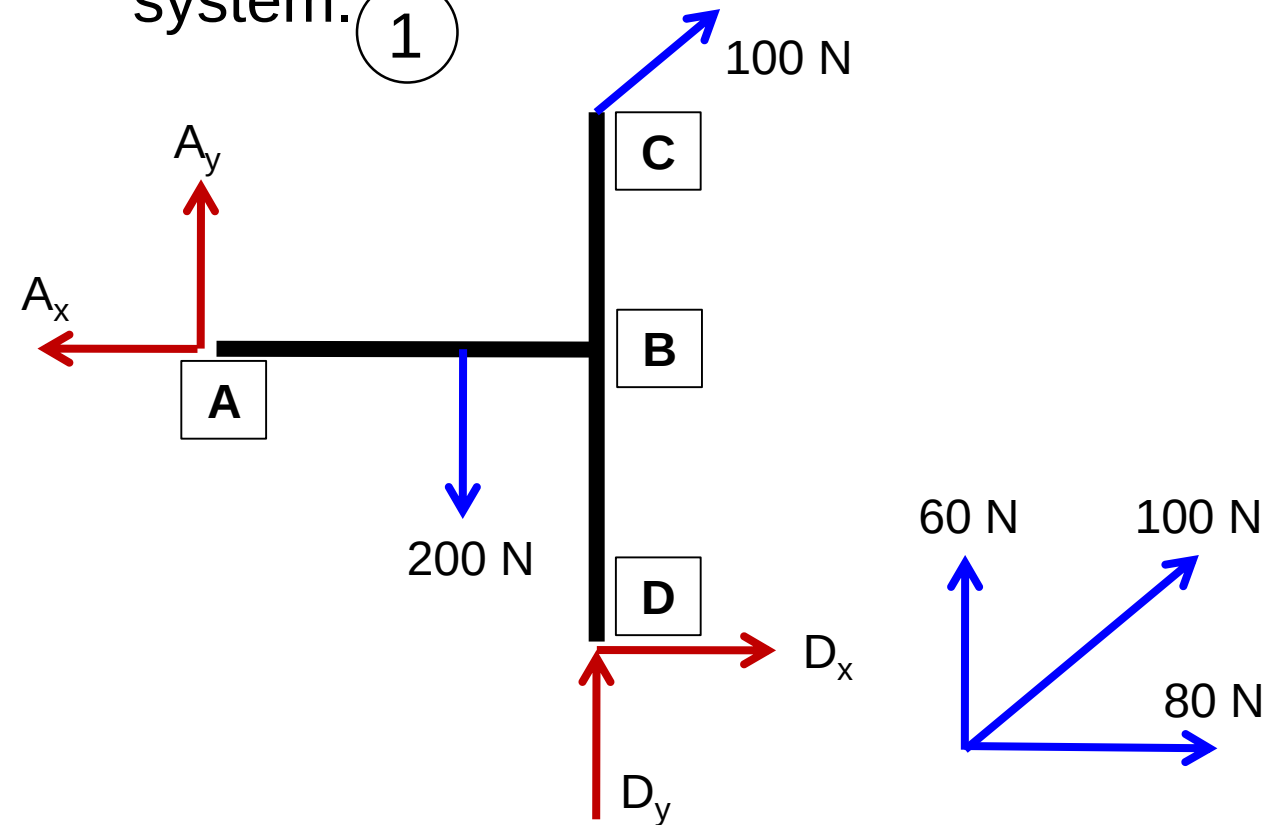
Example 1



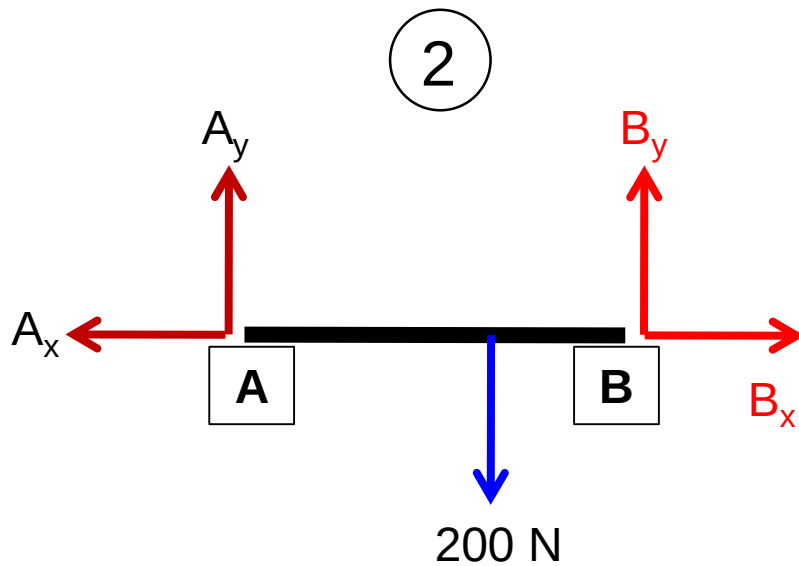
Find the reactions acting at point B

Step 1: FBD

- Identify all the reactions (external forces) and sketch the FBD for the full system. ①

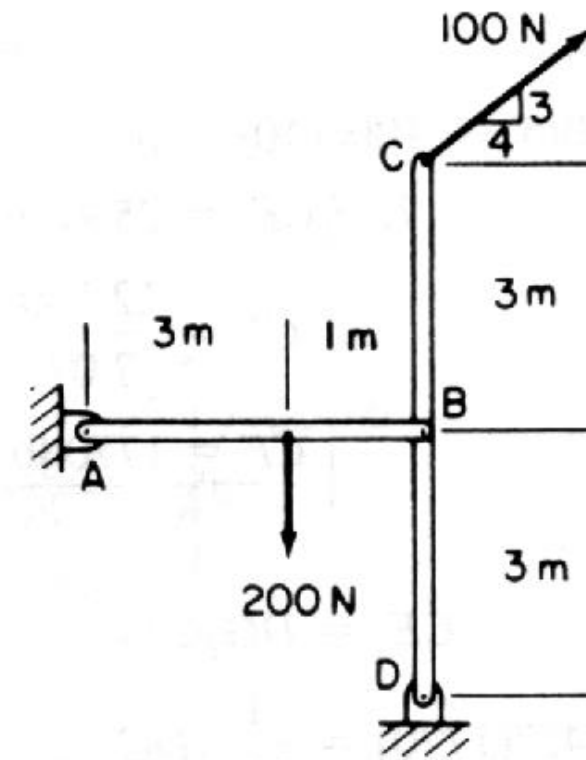


Example 1

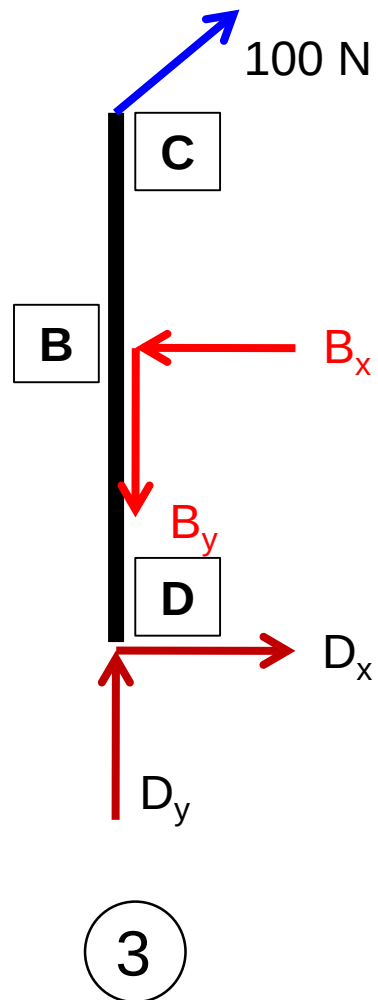


Step 2: FBD of individual member

- Sketch the FBD for the individual system. Note that Reactions at **point B** for each member **must compliment**

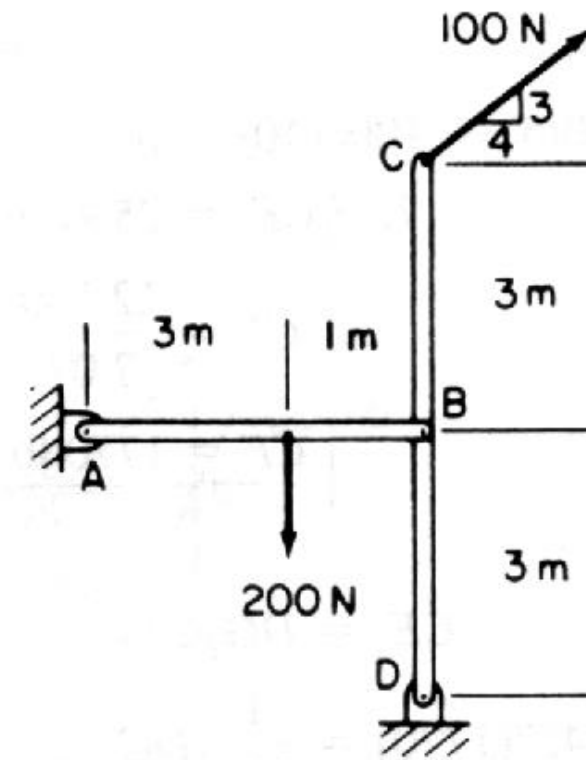


Example 1

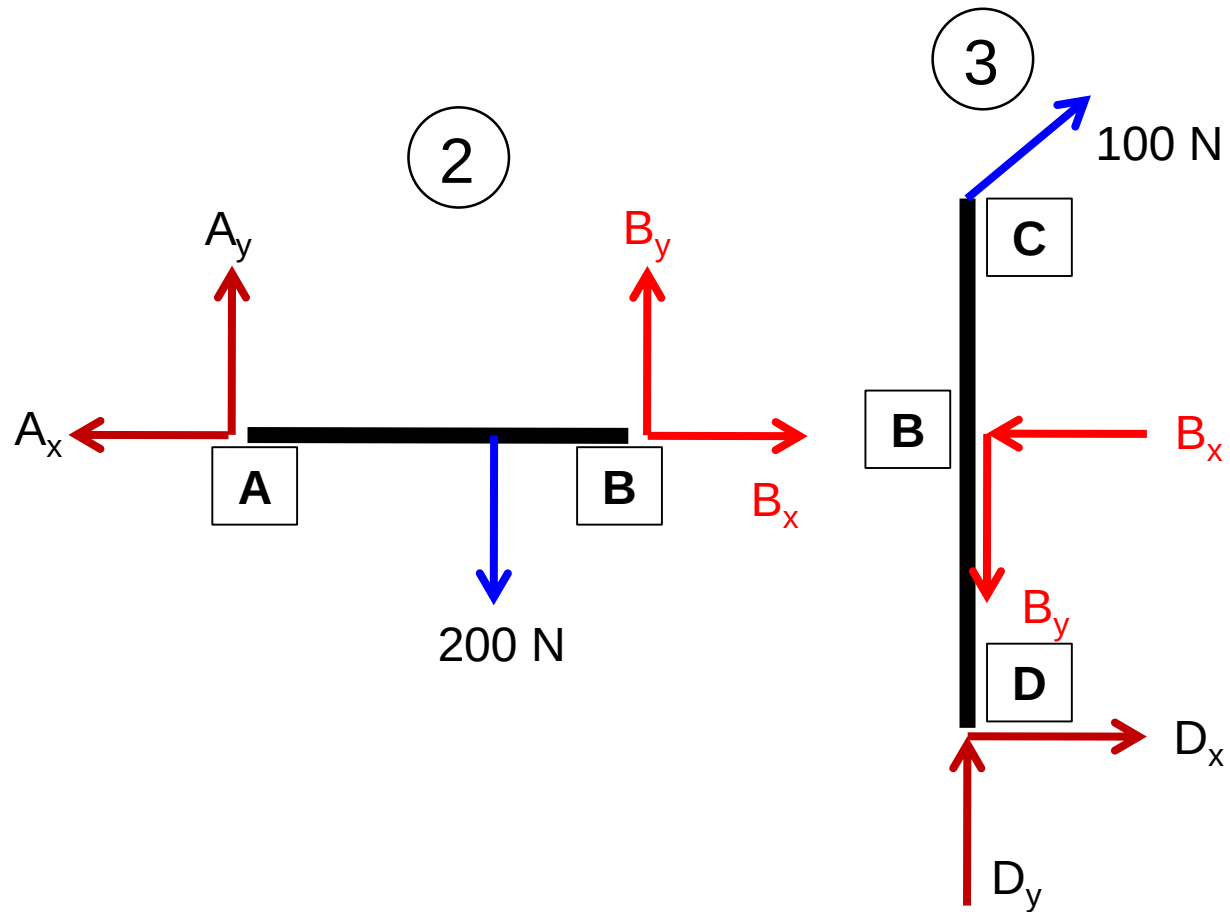


Step 2: FBD of individual member

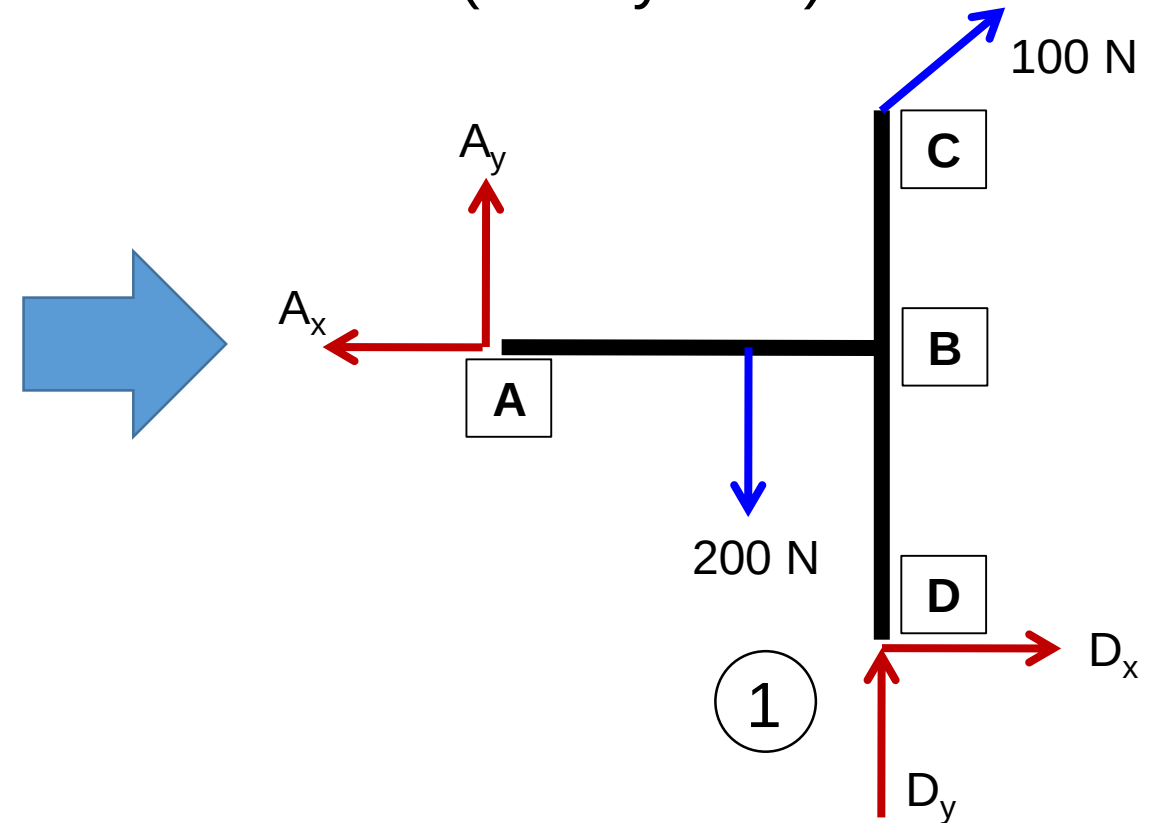
- Sketch the FBD for the individual system. Note that reactions at **point B** for each member **must compliment**



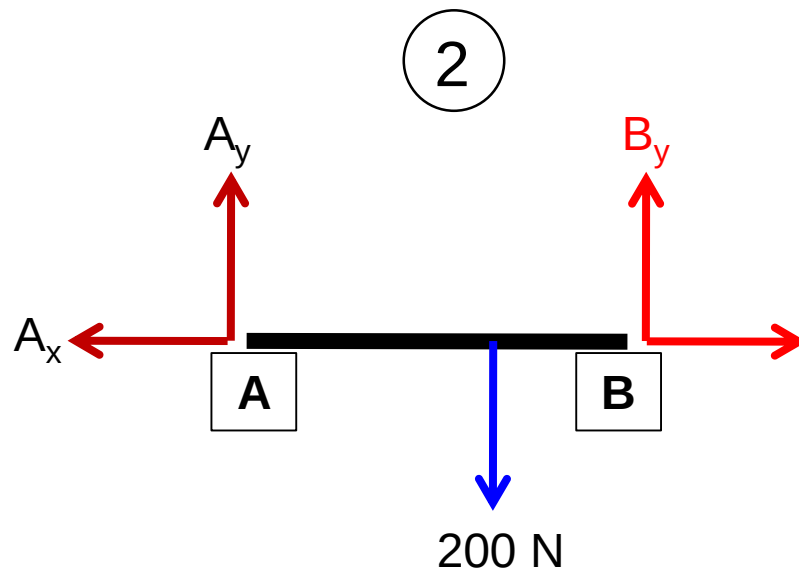
Example 1



- Note that reactions at **point B** for each member **must compliment**. If you add FBD 2 and FBD 3 together, it will become FBD 1 (Full system).



Example 1



Step 3: Static Equilibrium Analysis

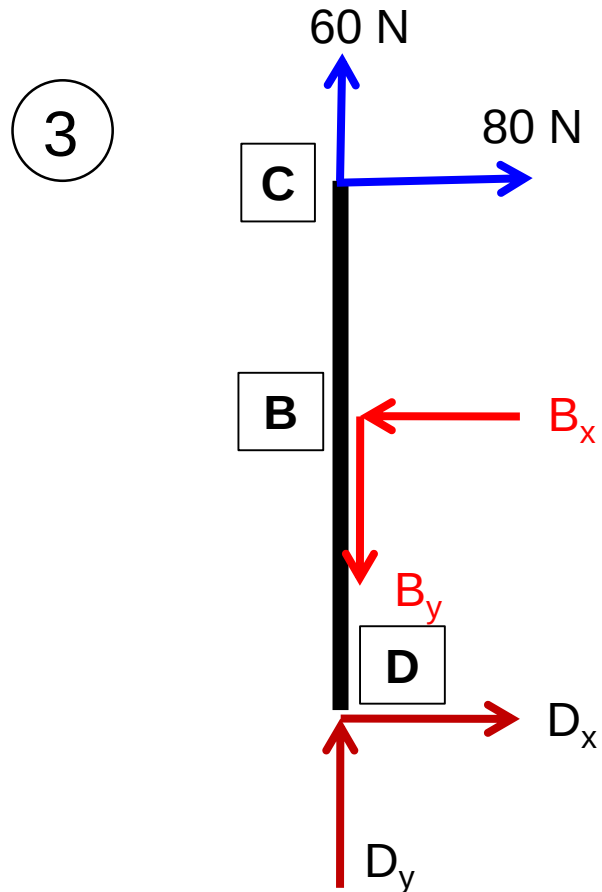
- Apply the three equations at respective FBD to solve for unknowns:
 $\sum F_x = 0$; $\sum F_y = 0$, $\sum M_z = 0$

Using $\sum M_A = 0$

$$3(200) - 4(B_y) = 0$$

$$B_y = \underline{150 \text{ N}}$$

Example 1



Step 3: Static Equilibrium Analysis

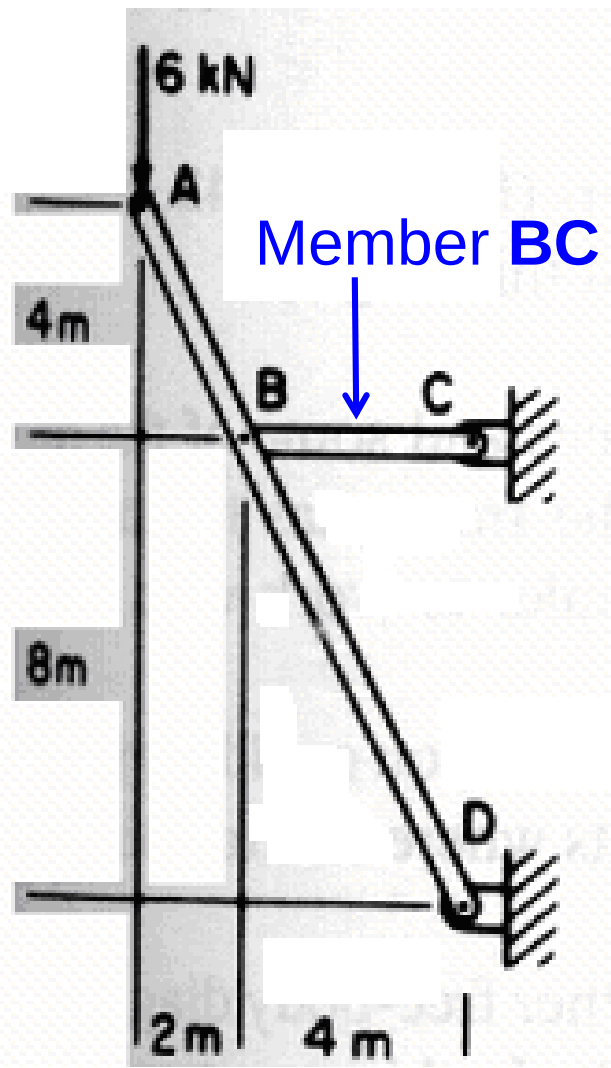
- Apply the three equations at respective FBD to solve for unknowns:
 $\sum F_x = 0$; $\sum F_y = 0$, $\sum M_z = 0$

Using $\sum M_D = 0$

$$6(80) - 3(B_x) = 0$$

$$B_x = \underline{160 \text{ N}}$$

Example 2

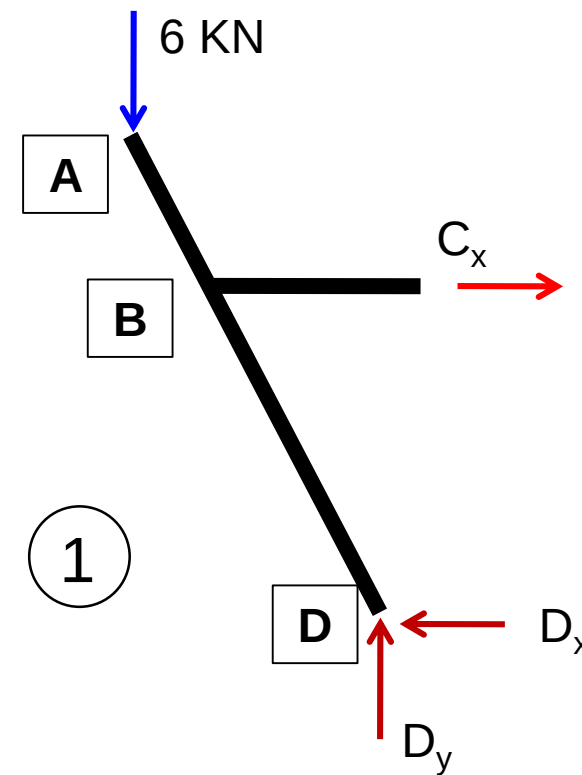


Member **BC** is a 2-force member

Find the reactions acting at point B

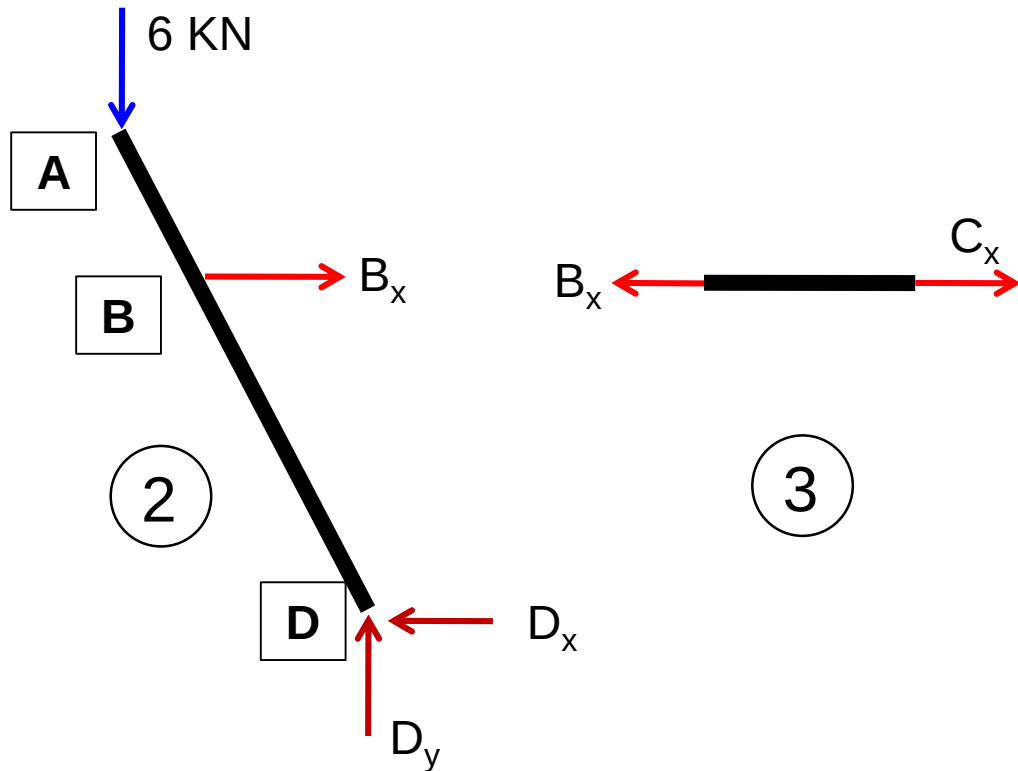
Step 1: FBD

- Identify all the reactions (external forces) and sketch the FBD for the full system.



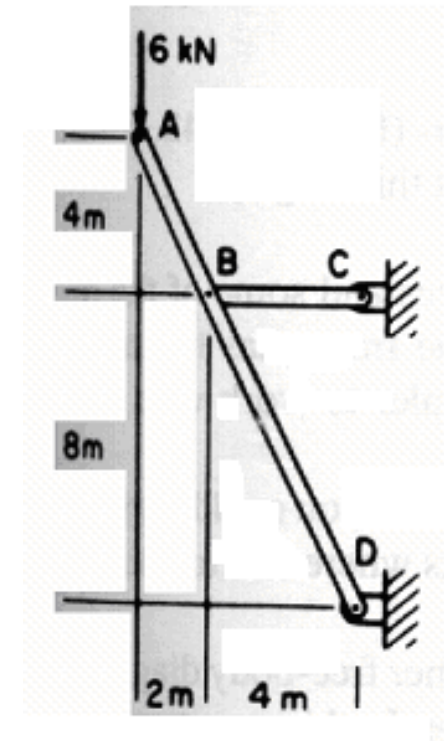
Note that only horizontal forces act on member BC because the member is positioned horizontally.

Example 2

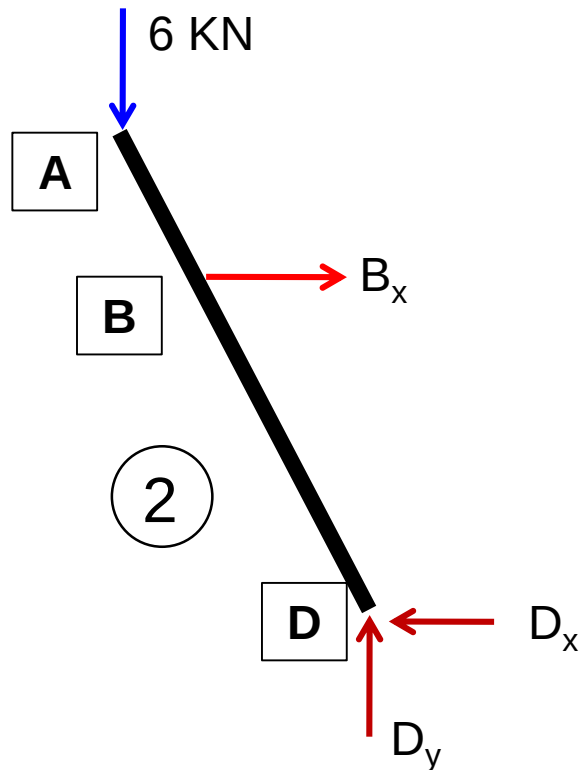


Step 2: FBD of individual member

- Sketch the FBD for the individual system. Note that reactions at **point B** for each member **must compliment**



Example 2



Step 3: Static Equilibrium Analysis

- Apply the three equations at respective FBD to solve for unknowns:
 $\sum F_x = 0$; $\sum F_y = 0$, $\sum M_z = 0$

Using $\sum F_y = 0$

$$D_y - 6k = 0 \rightarrow D_y = \underline{6 \text{ kN}}$$

Using $\sum M_B = 0$

$$(6)(2) + D_y(4) - D_x(8) = 0$$

$$D_x = \underline{4.5 \text{ kN}}$$

Question 1

1. Determine the horizontal and vertical components of the pin reactions at B and D of the frame shown in Figure 2.4.

Ans: $B_x = 7.5$ kips, $B_y = 0$, $D_x = 7.5$ kips, $D_y = 5$ kips

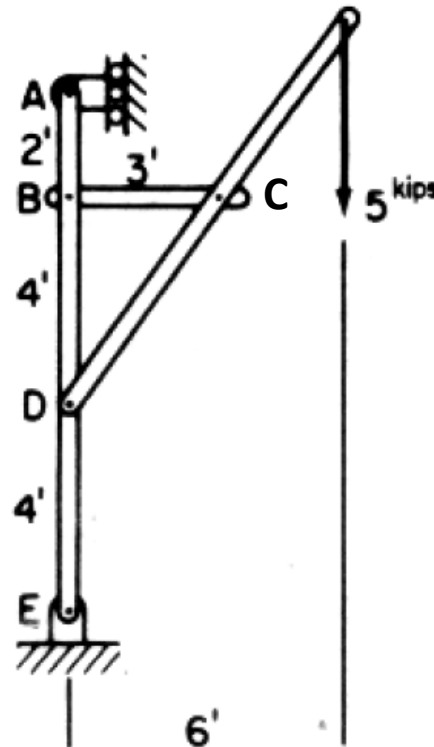


Figure 2.4

Question 1

1. Determine the horizontal and vertical components of the pin reactions at B and D of the frame shown in Figure 2.4.

Ans: $B_x = 7.5$ kips, $B_y = 0$, $D_x = 7.5$ kips, $D_y = 5$ kips

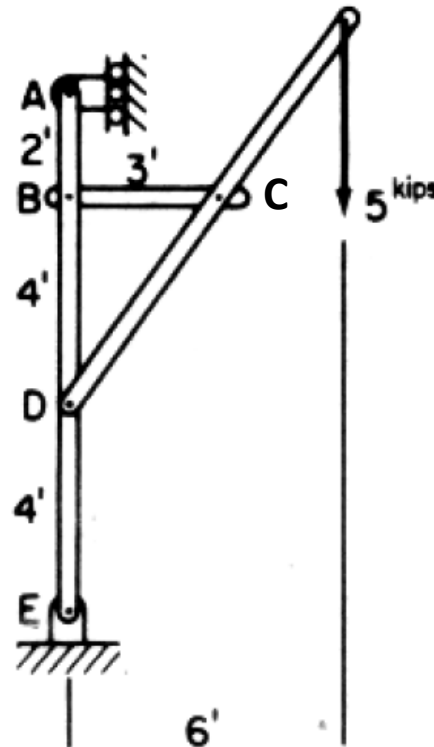
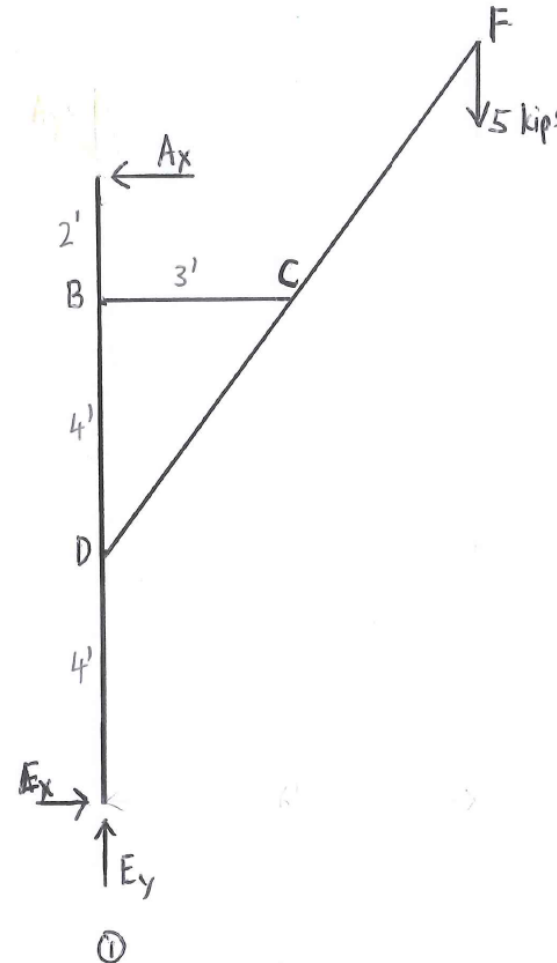


Figure 2.4

Question 1

Step 1 & 2: FBD and FBD of individual member

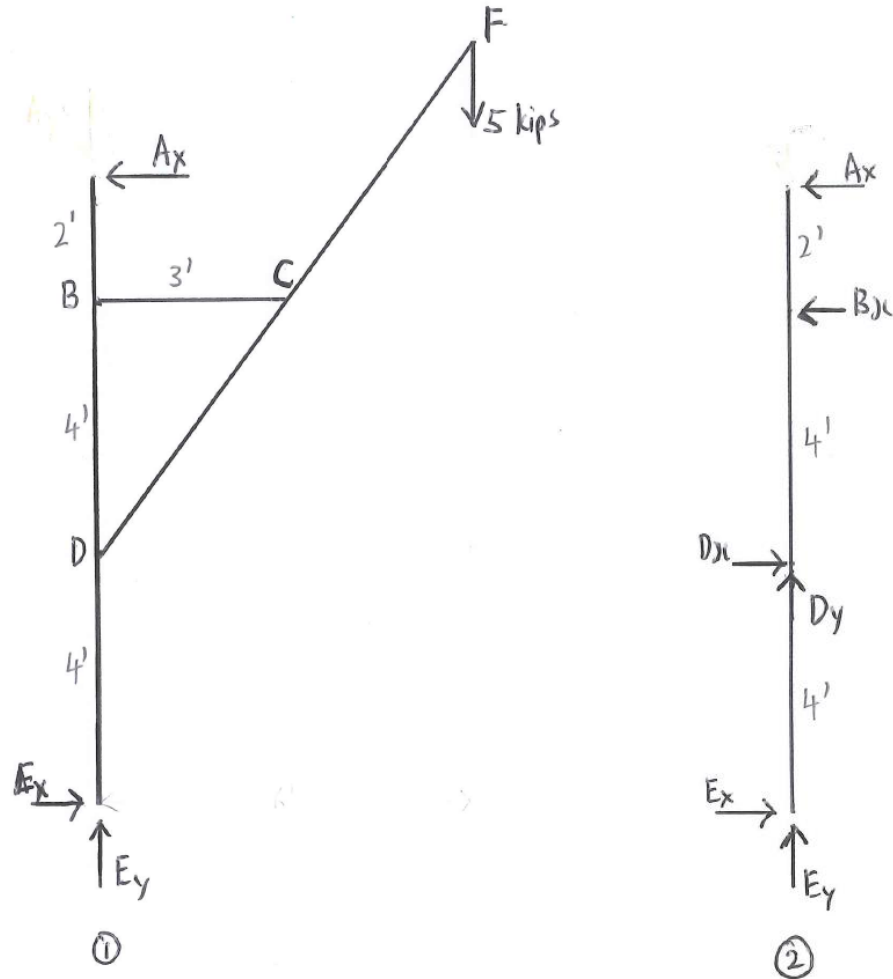
- Identify all the reactions (external forces) and sketch the FBD for the full system.



Question 1

Step 1 & 2: FBD and FBD of individual member

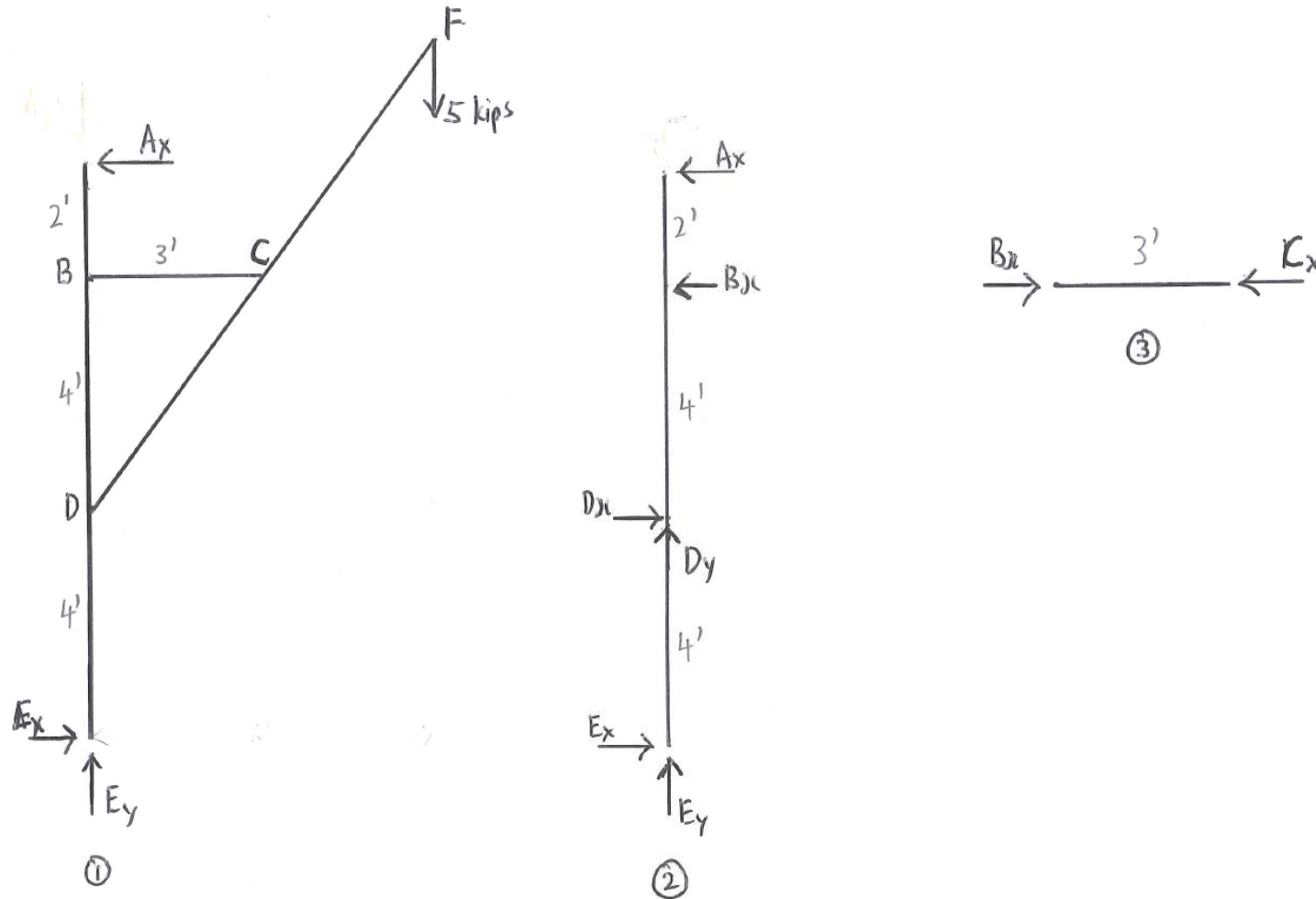
- Identify all the reactions (external forces) and sketch the FBD for the full system.



Question 1

Step 1 & 2: FBD and FBD of individual member

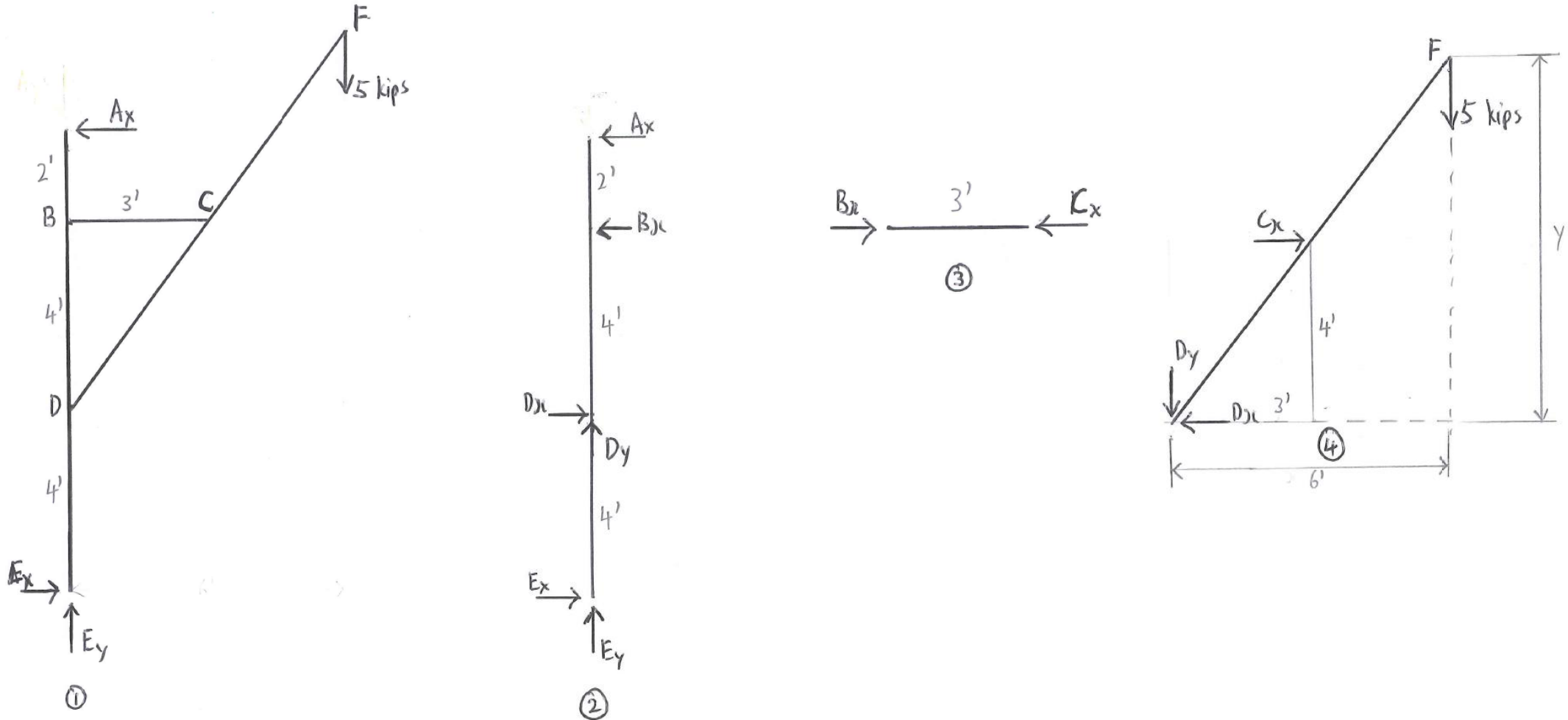
- Identify all the reactions (external forces) and sketch the FBD for the full system.



Question 1

Step 1 & 2: FBD and FBD of individual member

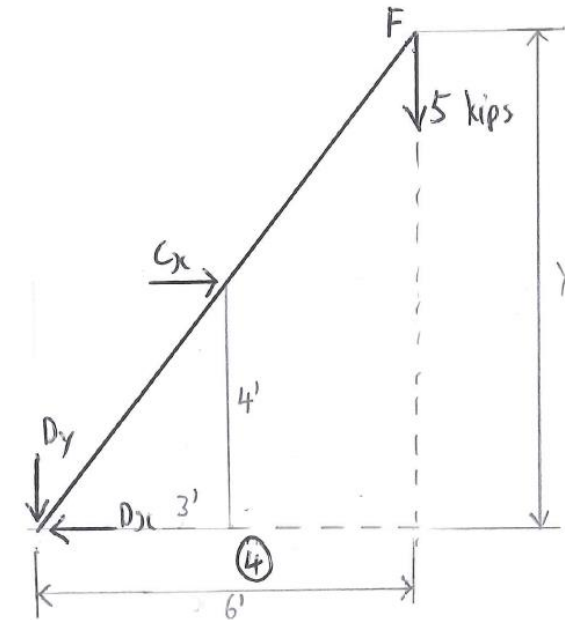
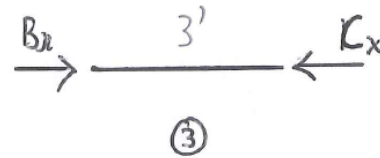
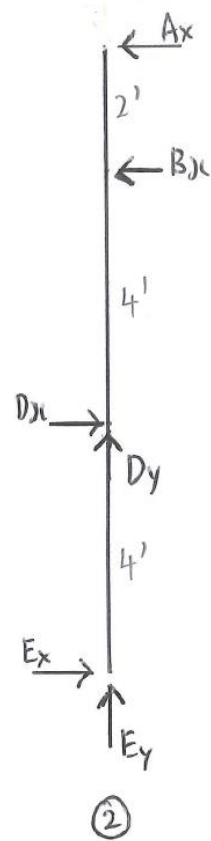
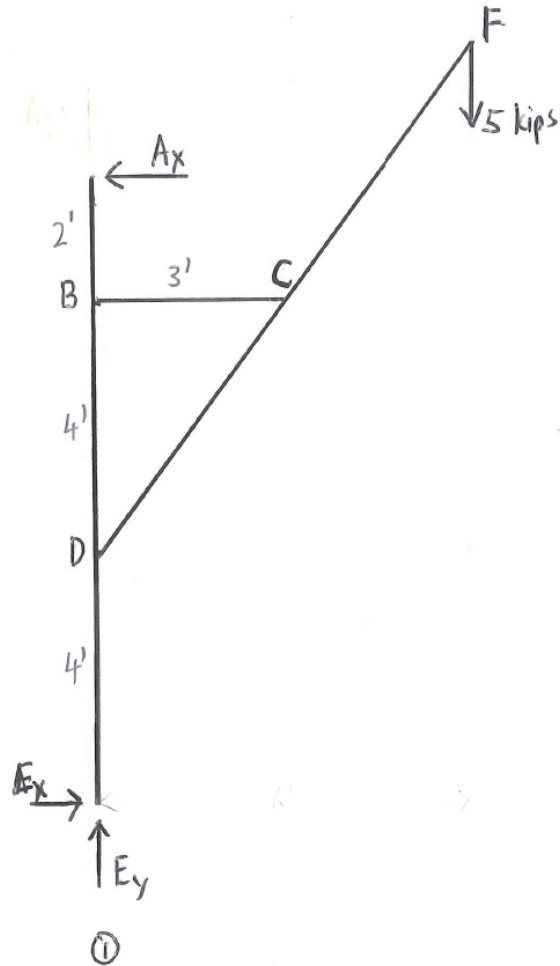
- Identify all the reactions (external forces) and sketch the FBD for the full system.



Question 1

Step 1 & 2: FBD and FBD of individual member

- Identify all the reactions (external forces) and sketch the FBD for the full system.

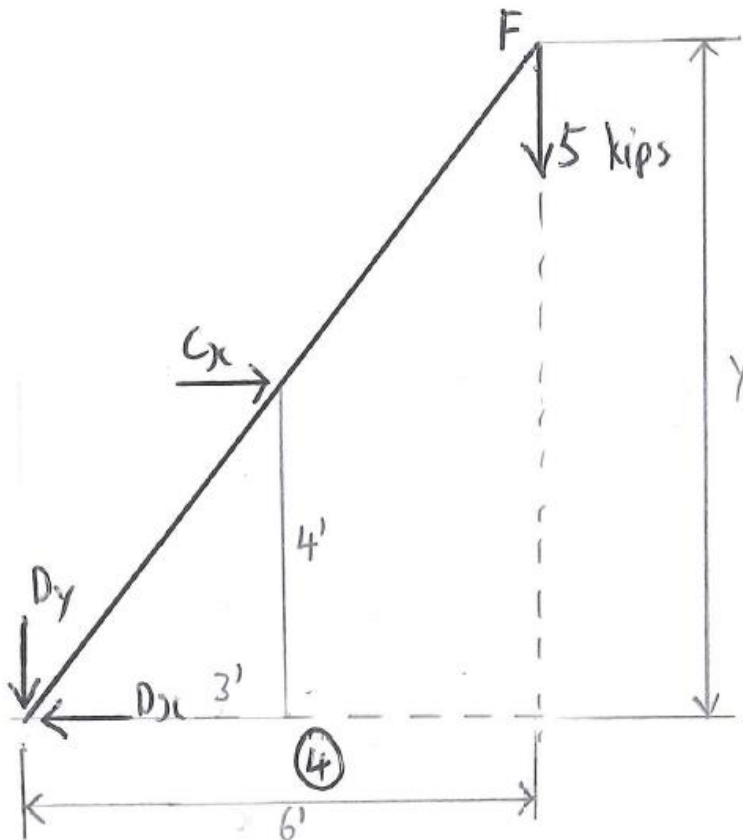


By similar triangle, $\frac{y}{6'} = \frac{4'}{3'}$
 $\therefore y = 2 \times 4'$
 $= \underline{8'}$

Question 1

Step 3: Static Equilibrium Analysis

- Apply the three equations at respective FBD to solve for unknowns:
 $\sum F_x = 0$; $\sum F_y = 0$, $\sum M_z = 0$



FBD ④:

$$\sum M_D = 0:$$

$$(4)(C_x) + (6)(5) = 0$$

$$C_x = -30/4$$

$$= \underline{\underline{-7.5 \text{ kips}}}$$

$$\sum F_x = 0$$

$$C_x = D_x$$

$$D_x = \underline{\underline{-7.5 \text{ kips}}}$$

$$\sum F_y = 0:$$

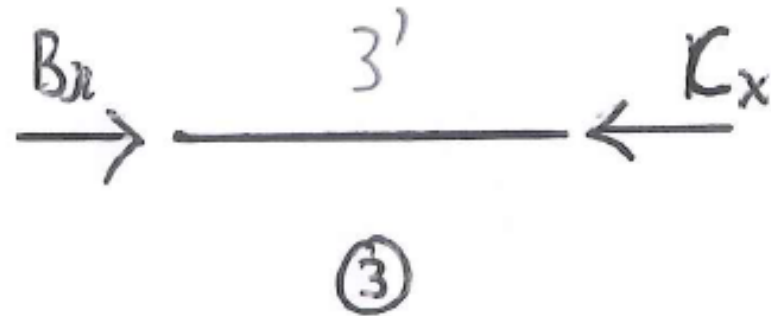
$$-D_y - 5 = 0$$

$$D_y = \underline{\underline{5 \text{ kips}}}$$

Question 1

Step 3: Static Equilibrium Analysis

- Apply the three equations at respective FBD to solve for unknowns:
 $\sum F_x = 0$; $\sum F_y = 0$, $\sum M_z = 0$



FBD ③:

Member BC is a two-force member.

$$\sum F_x = 0:$$

$$B_x = C_x = \underline{\underline{-7.5 \text{ kips}}}$$

Question 1

Step 3: Static Equilibrium Analysis

- Apply the three equations at respective FBD to solve for unknowns:
 $\sum F_x = 0$; $\sum F_y = 0$, $\sum M_z = 0$

FBD ①:

$$\sum F_y = 0:$$

$$E_y - 5 = 0$$

$$E_y = \underline{\underline{5 \text{ kips}}}$$

$$\sum M_B = 0:$$

$$(6)(5) - (10)(A_x) = 0$$

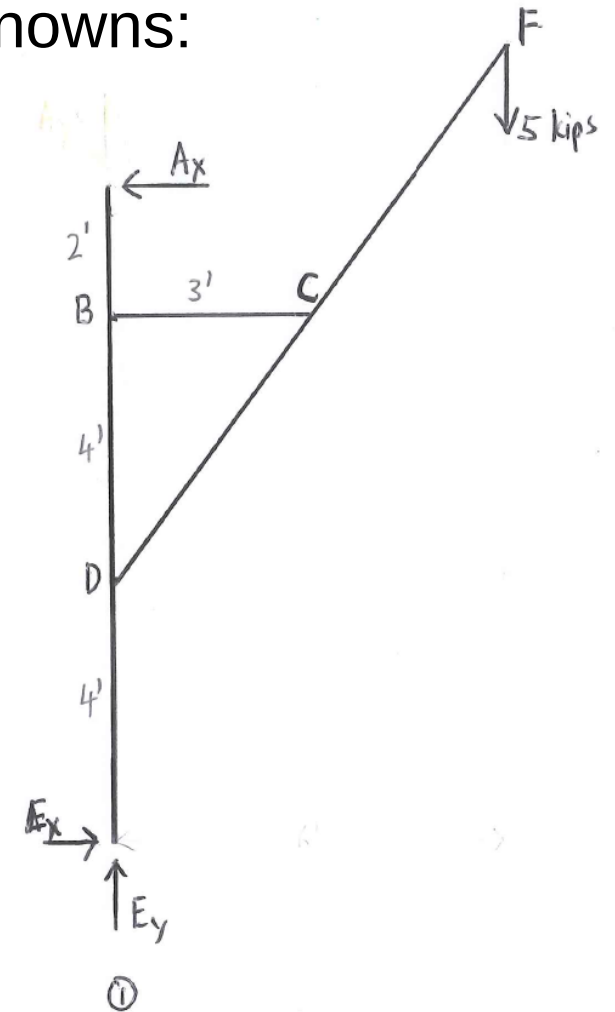
$$A_x = 30/10$$

$$= \underline{\underline{3 \text{ kips}}}$$

$$\sum F_x = 0$$

$$E_x = A_x$$

$$E_x = \underline{\underline{3 \text{ kips}}}$$



Question 2

2. The frame in Figure 2.5 is held by cable HG when loaded as shown. Determine the horizontal and vertical components of the pin reactions at B and C.

Ans: $B_x = 3000 \text{ lb}$; $B_y = 4200 \text{ lb}$; $C_x = 3000 \text{ lb}$; $C_y = 3000 \text{ lb}$

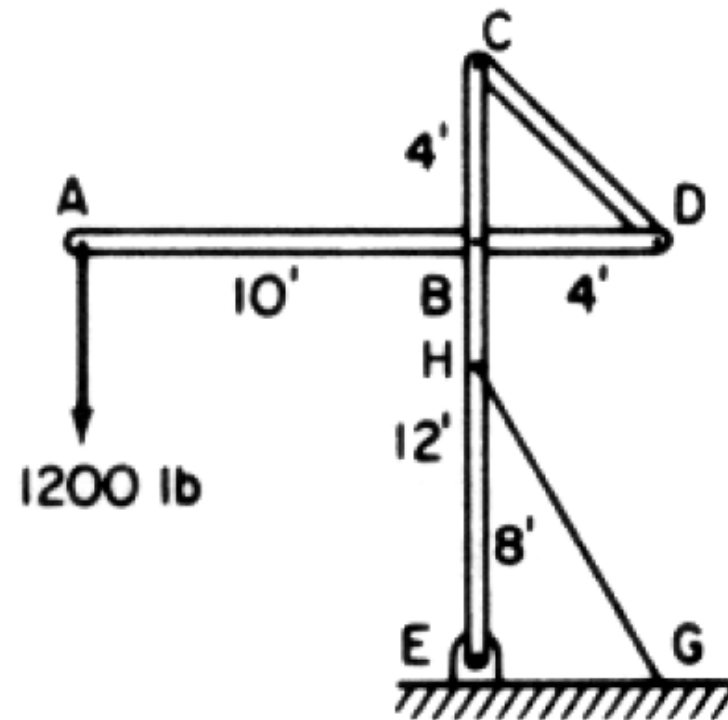


Figure 2.5

Question 2

2. The frame in Figure 2.5 is held by cable HG when loaded as shown. Determine the horizontal and vertical components of the pin reactions at B and C.

Ans: $B_x = 3000 \text{ lb}$; $B_y = 4200 \text{ lb}$; $C_x = 3000 \text{ lb}$; $C_y = 3000 \text{ lb}$

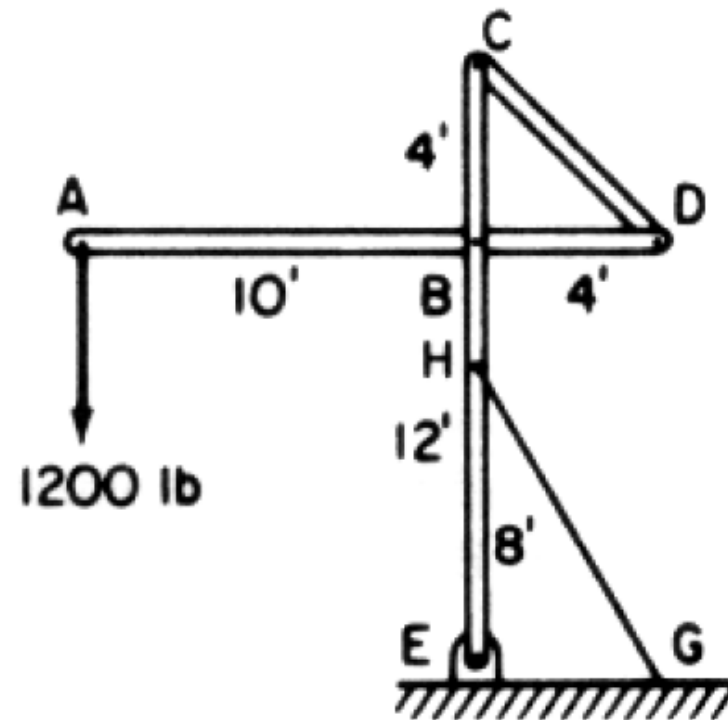
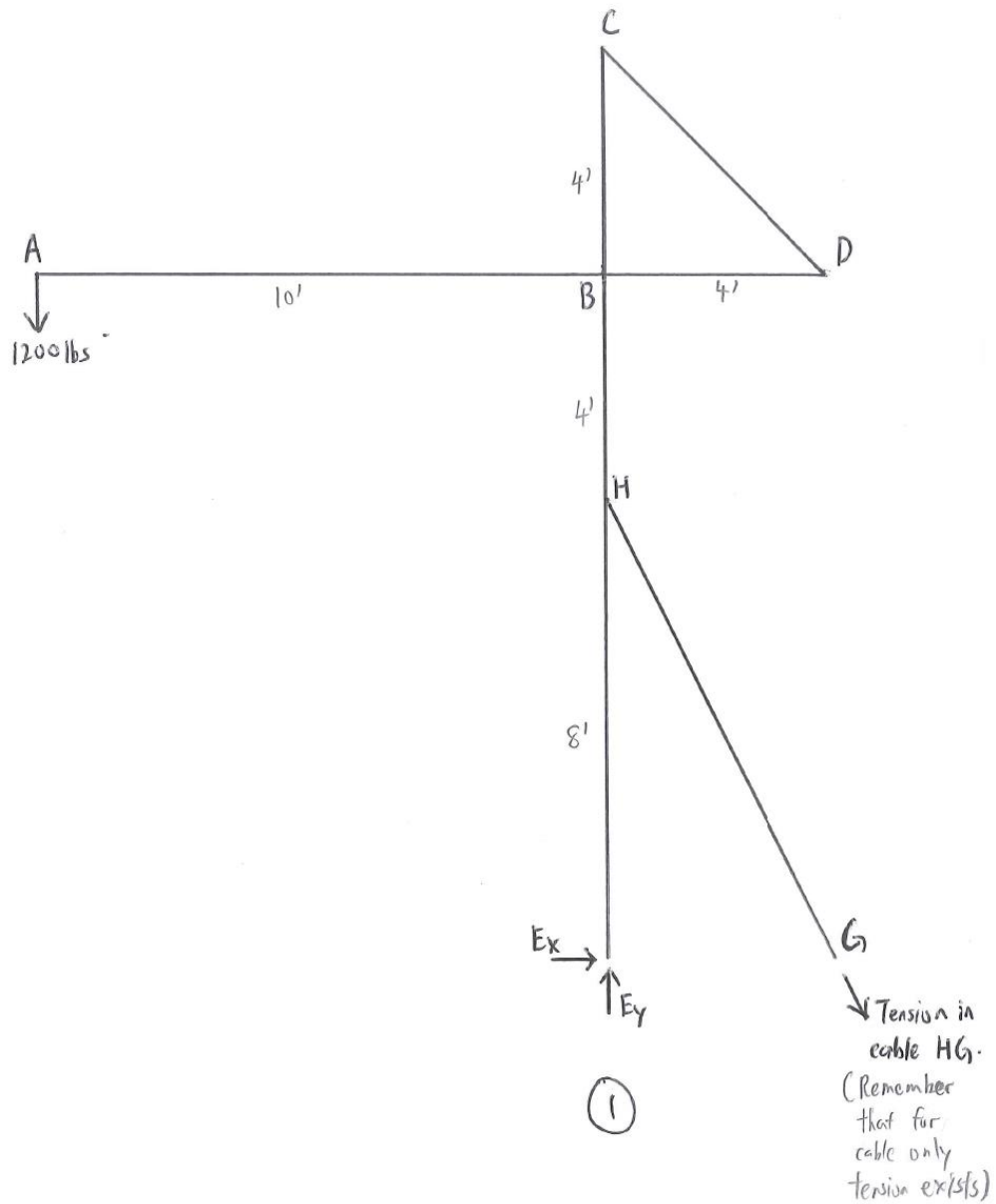
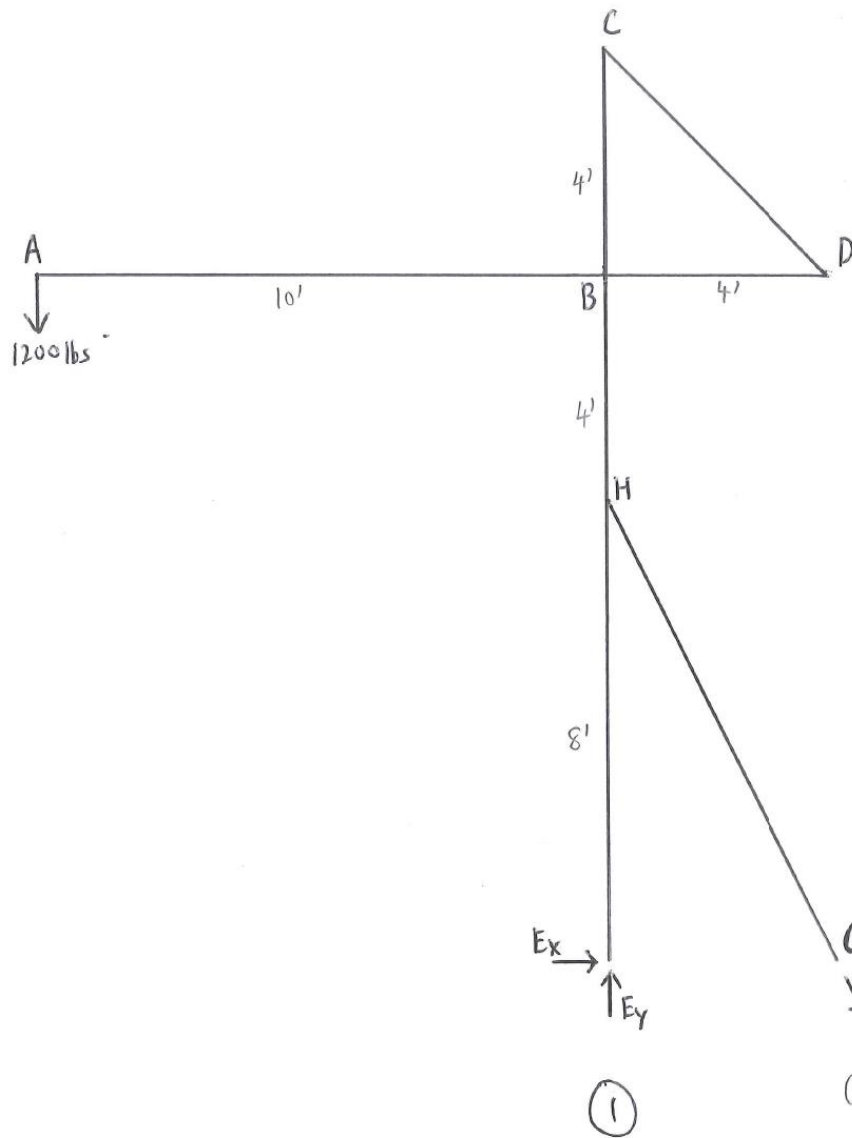


Figure 2.5

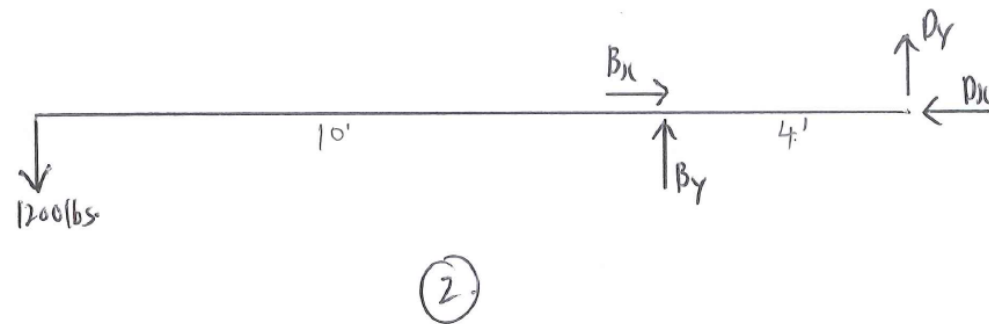
Question 2



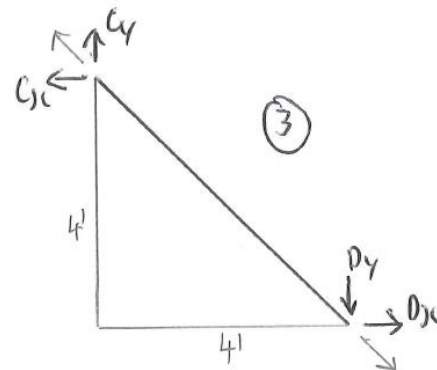
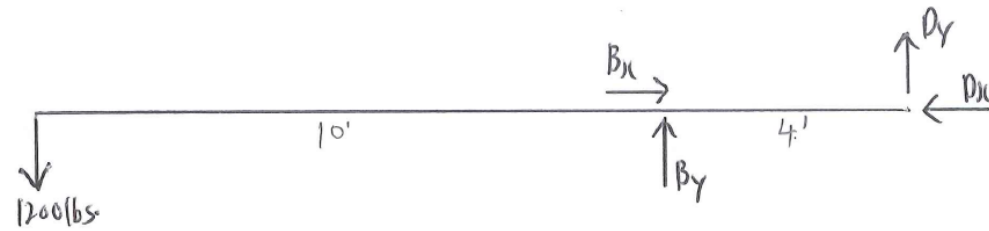
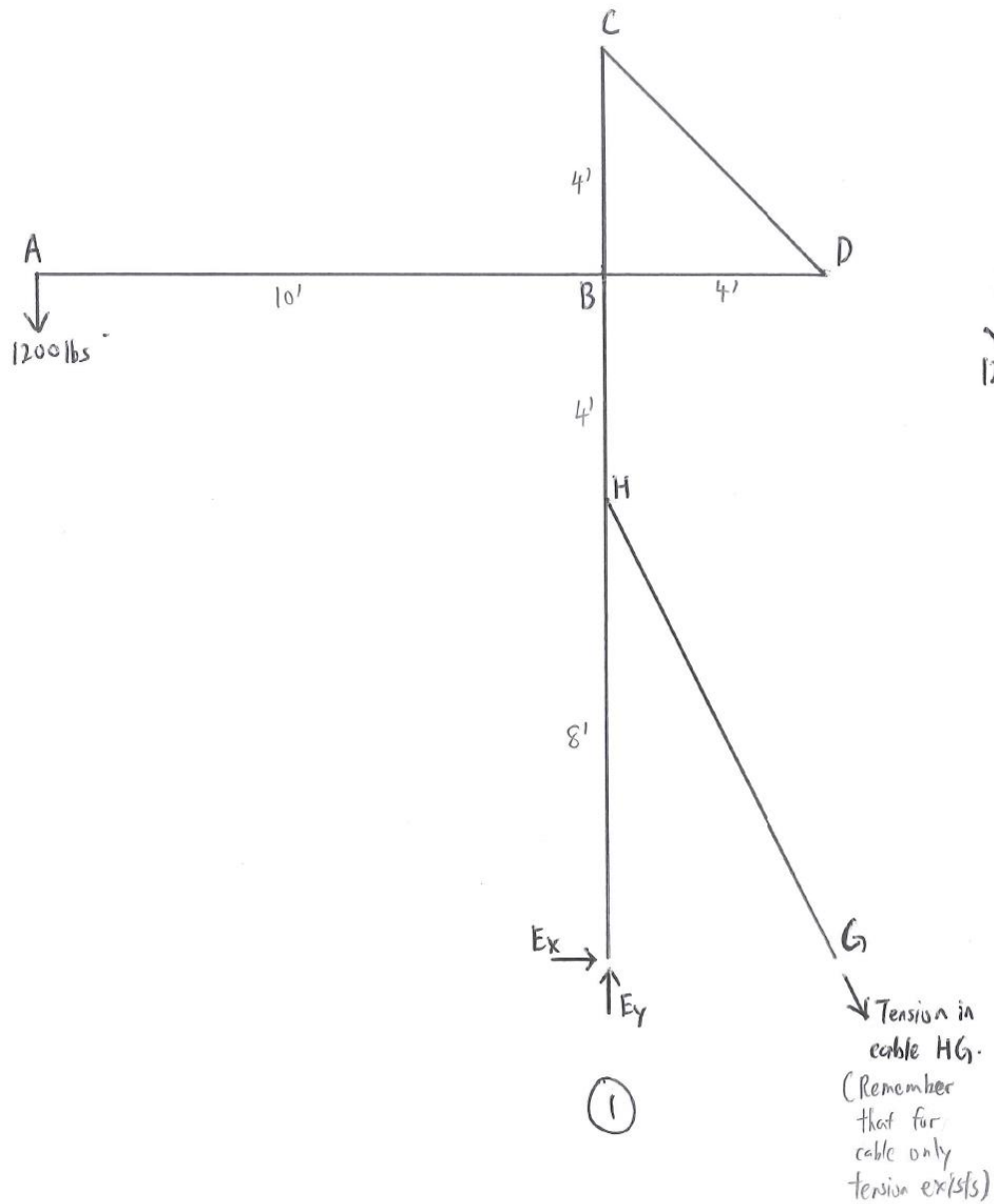
Question 2



↓ Tension in cable HG.
(Remember that for cable only tension exists)

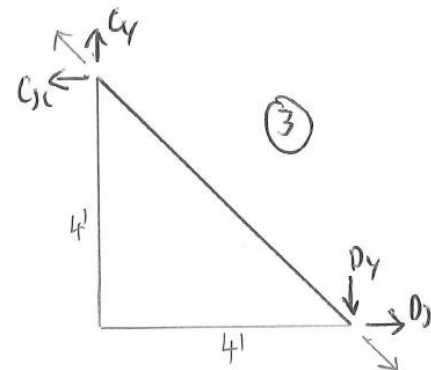
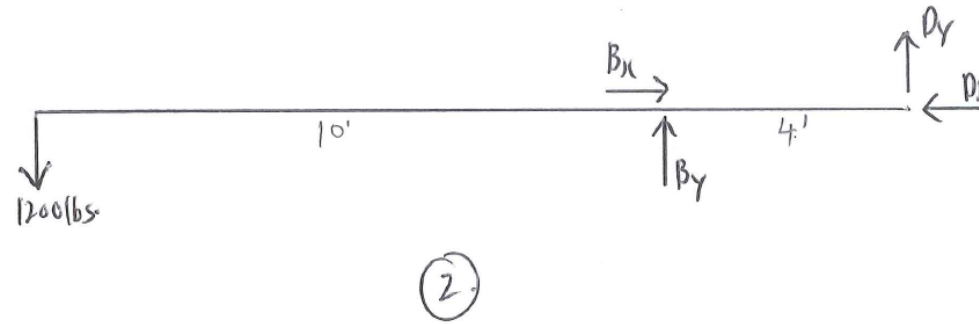
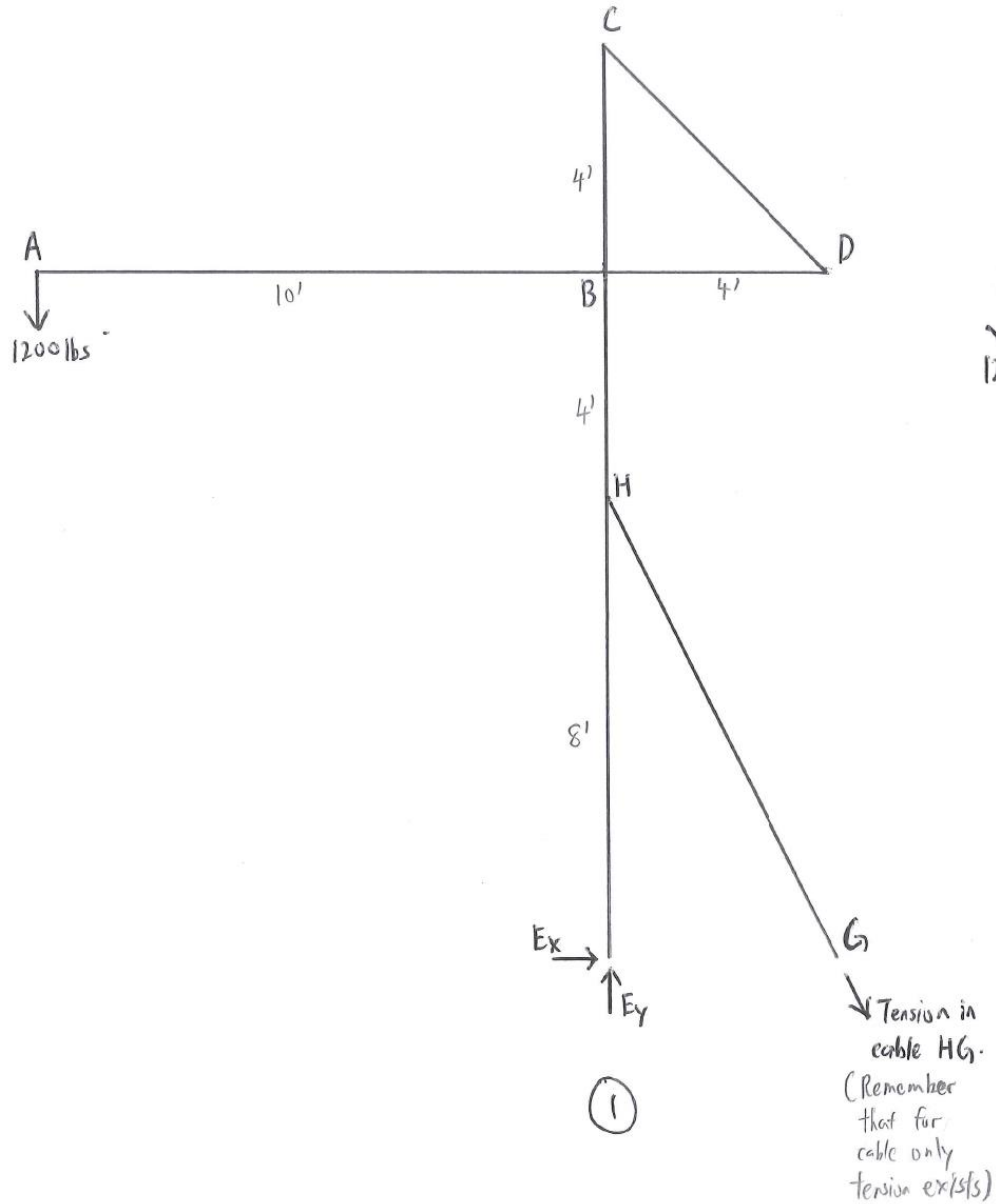


Question 2

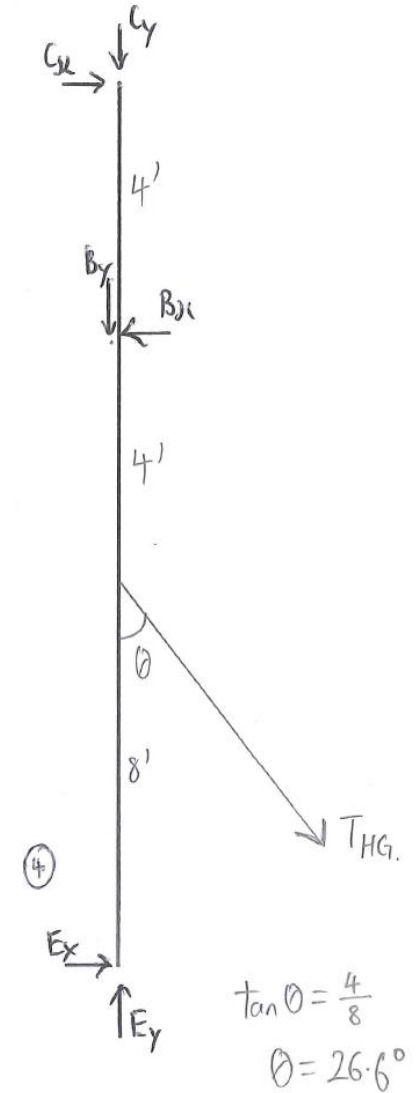


Remember for a two-force member, always assume it to be in tension or compression

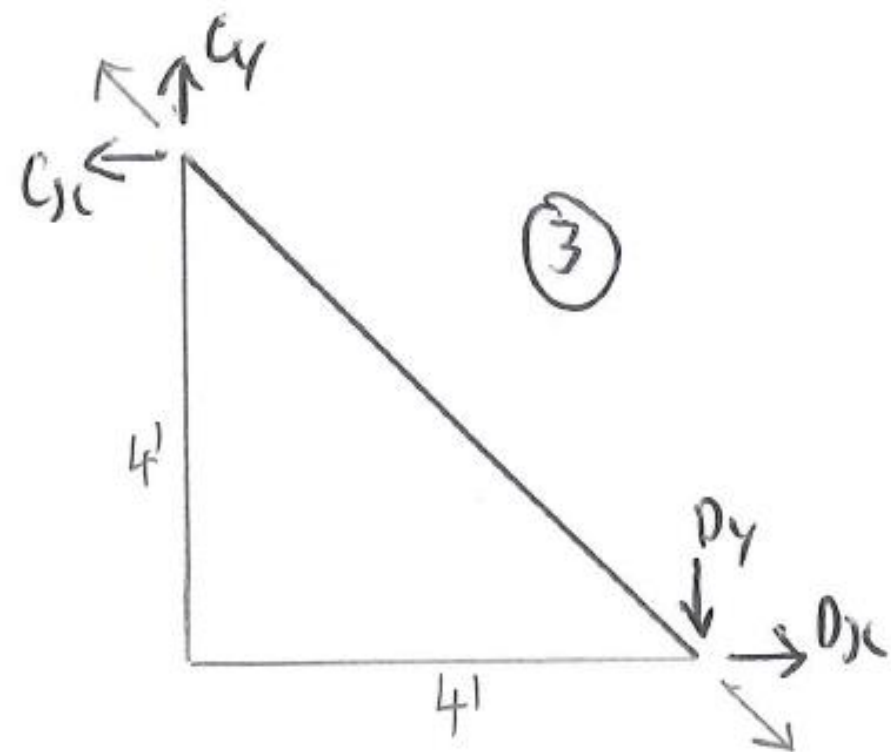
2 Question 2



Remember for a two-force member, always assume it to be in tension or compression



Question 2



Remember for a two-force member, always assume it to be in tension or compression

FBD (3)

$$\sum M_C = 0$$

$$4 D_y - 4 D_x = 0$$

$$4 D_y = 4 D_x$$

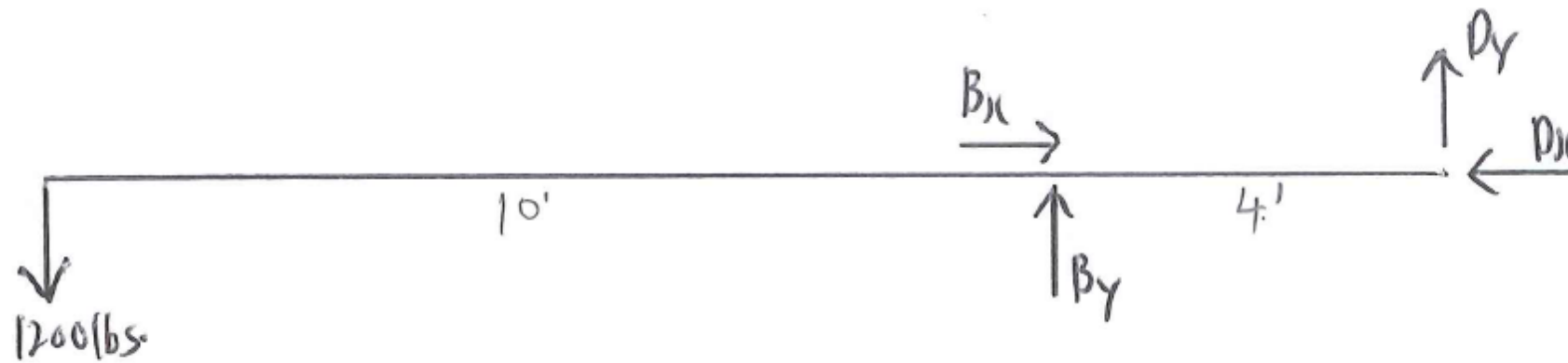
$$\frac{D_y}{D_x} = \frac{4}{4}$$

$$\Rightarrow D_y = D_x$$

$$\sum F_x = 0 \Rightarrow C_x = D_x$$

$$\sum F_y = 0 \Rightarrow C_y = D_y$$

Question 2



FBD ②

$$\sum M_B = 0$$

$$(1200)(10) + (4)(D_y) = 0$$

$$D_y = -12000/4$$

$$= -3000 \text{ lbs/ft}$$

$$\Rightarrow D_x = -3000 \text{ lbs/ft}$$

$$\Rightarrow C_x = -3000 \text{ lbs/ft}$$

$$\Rightarrow C_y = -3000 \text{ lbs/ft}$$

②

$$\sum F_y = 0$$

$$B_y + D_y - 1200 = 0$$

$$B_y = 1200 + 3000$$

$$= 4200 \text{ lbs/ft}$$

$$\sum F_x = 0$$

$$B_x - D_x = 0$$

$$B_x = D_x$$

$$= -3000 \text{ lbs/ft}$$

Question 3

3. Determine the horizontal and vertical components of the pin reaction at D of the system shown in Figure 2.6. Assume the contact at G is with a frictionless surface.

Ans: $D_x = 76.8 \text{ kN}$ $D_y = 114 \text{ kN}$

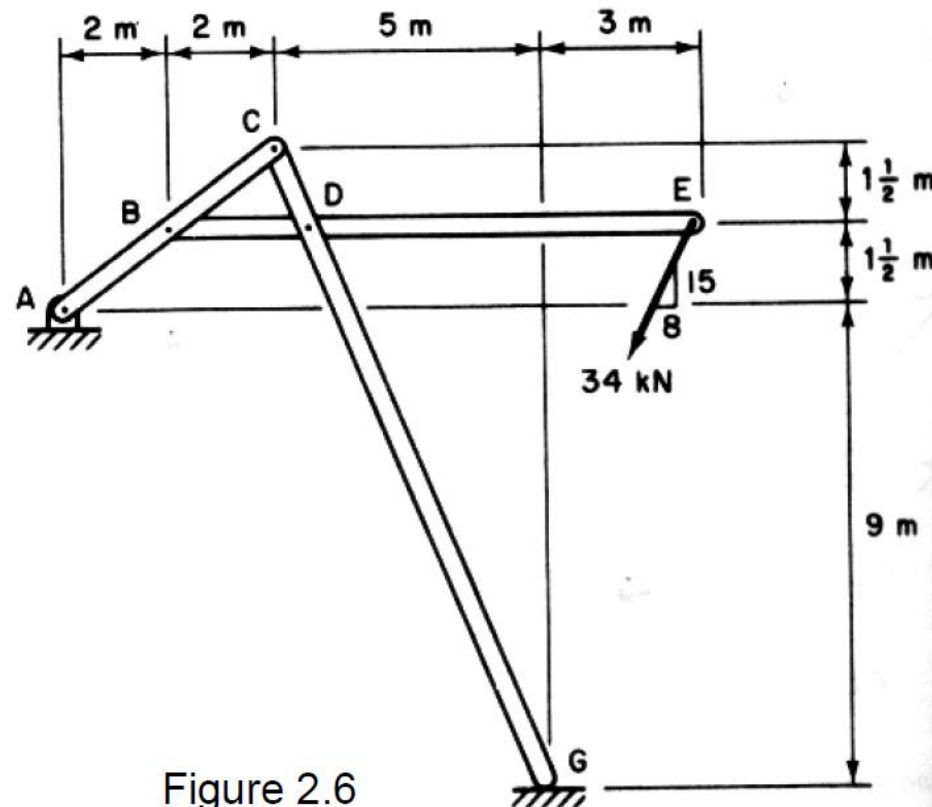


Figure 2.6

Question 3

3. Determine the horizontal and vertical components of the pin reaction at D of the system shown in Figure 2.6. Assume the contact at G is with a frictionless surface.

Ans: $D_x = 76.8 \text{ kN}$ $D_y = 114 \text{ kN}$

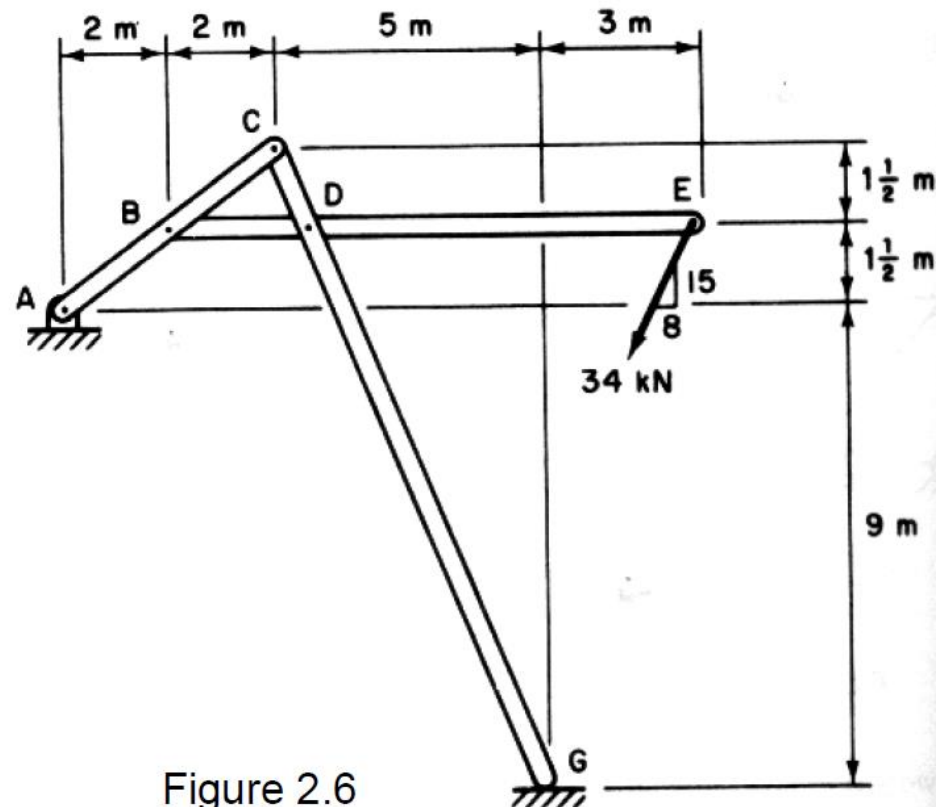
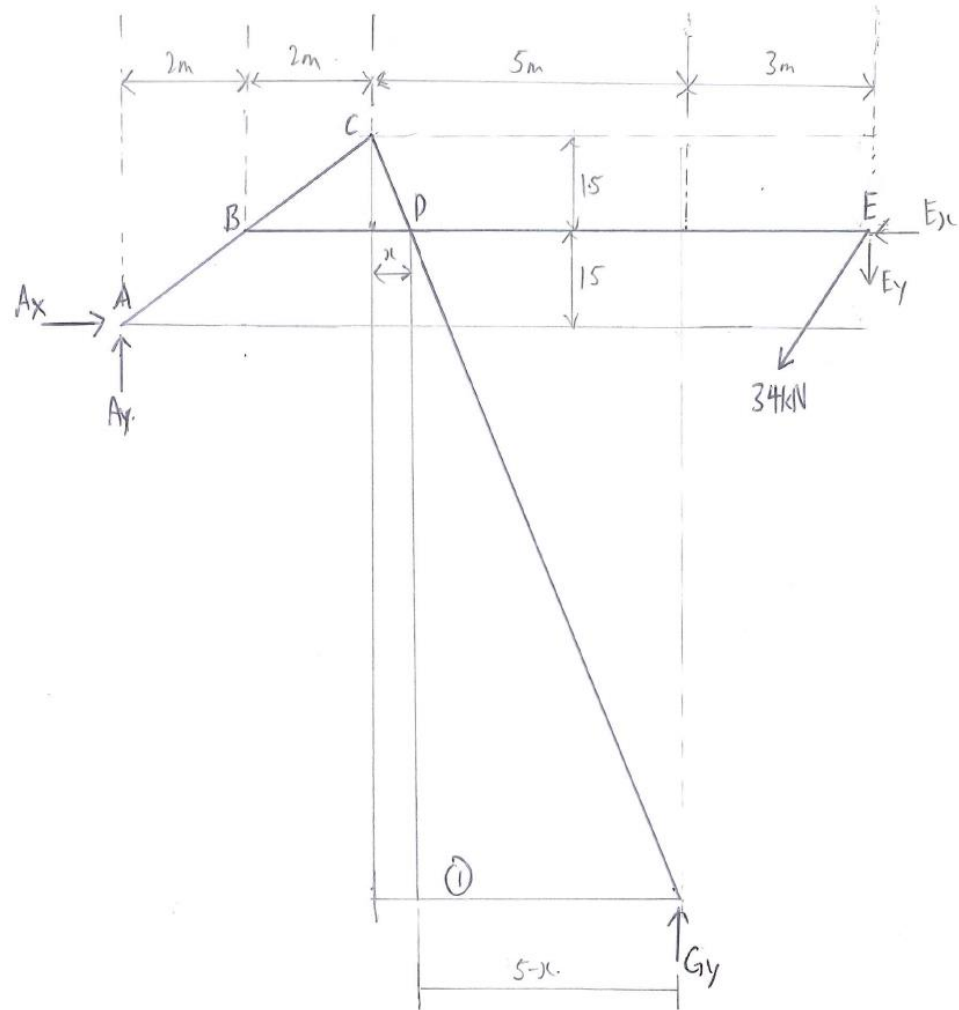
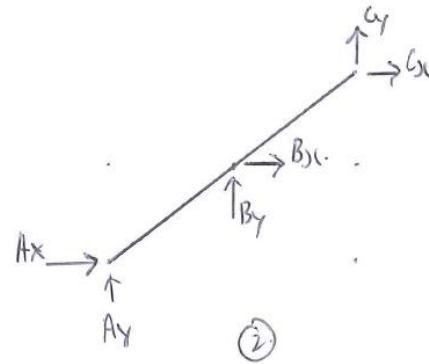
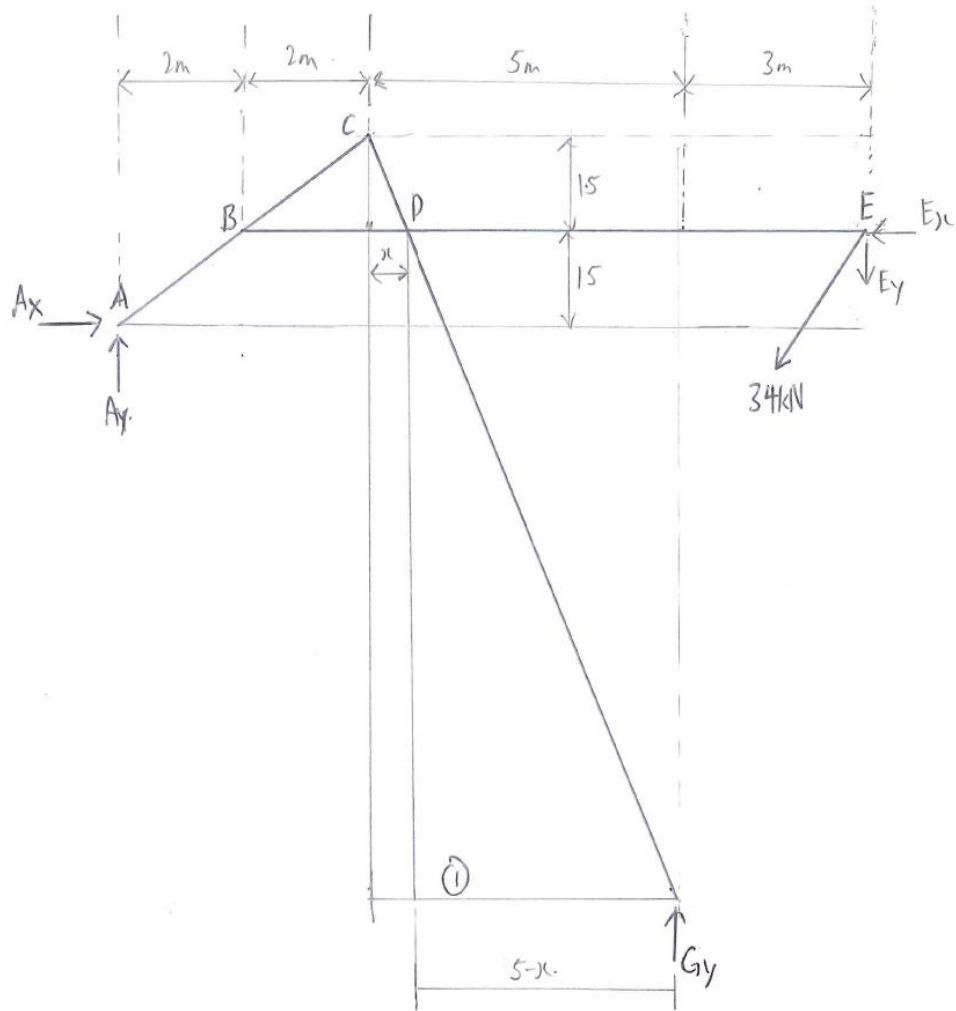


Figure 2.6

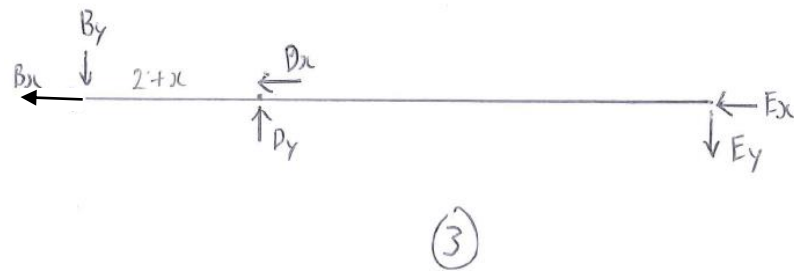
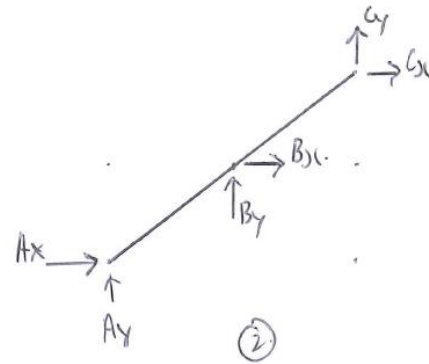
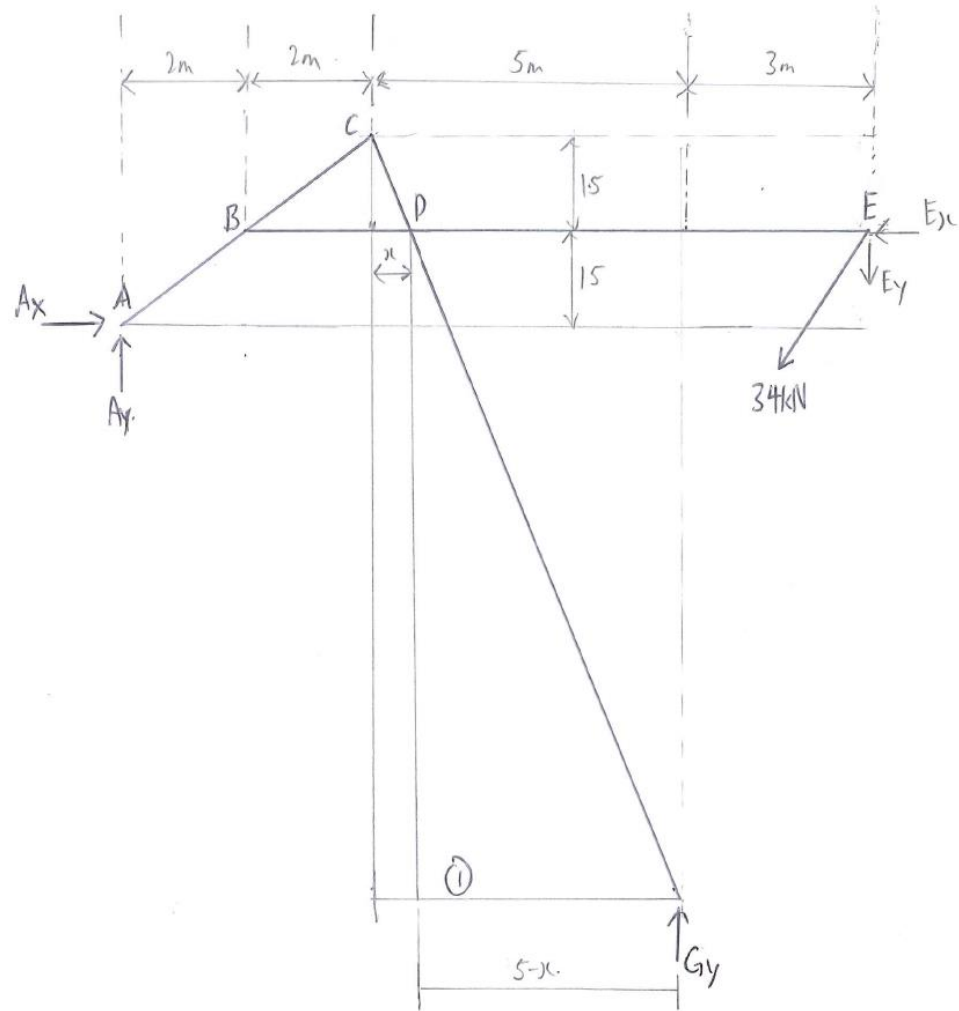
Question 3



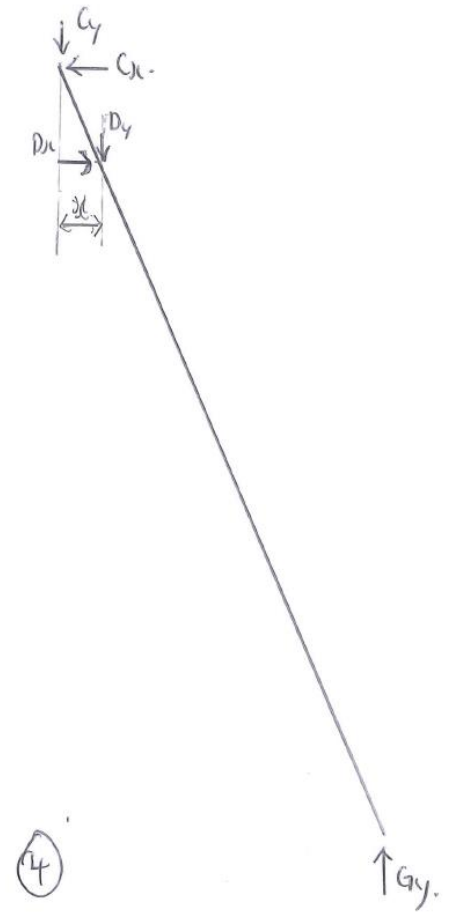
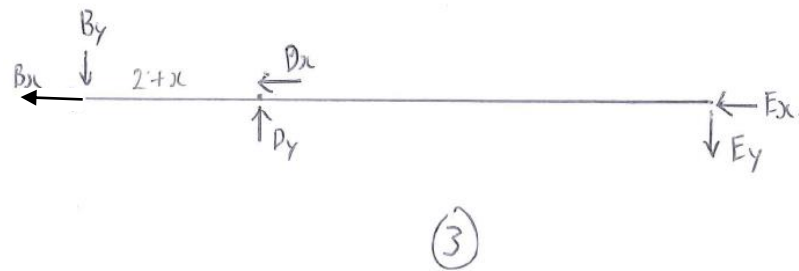
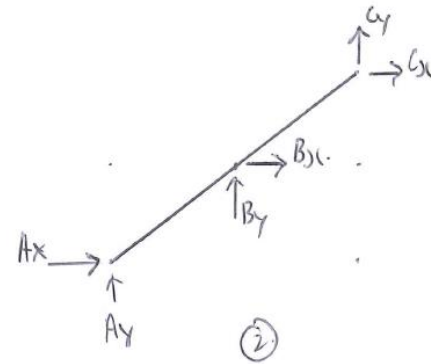
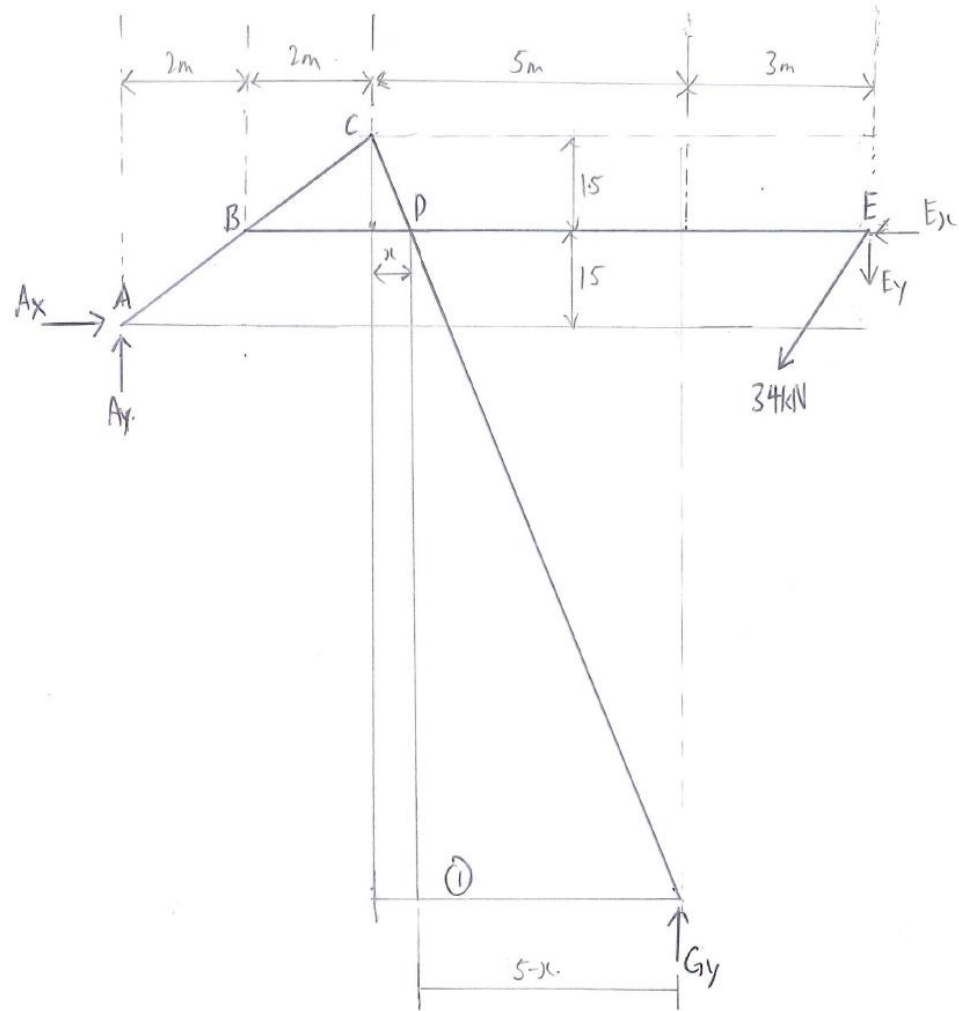
Question 3



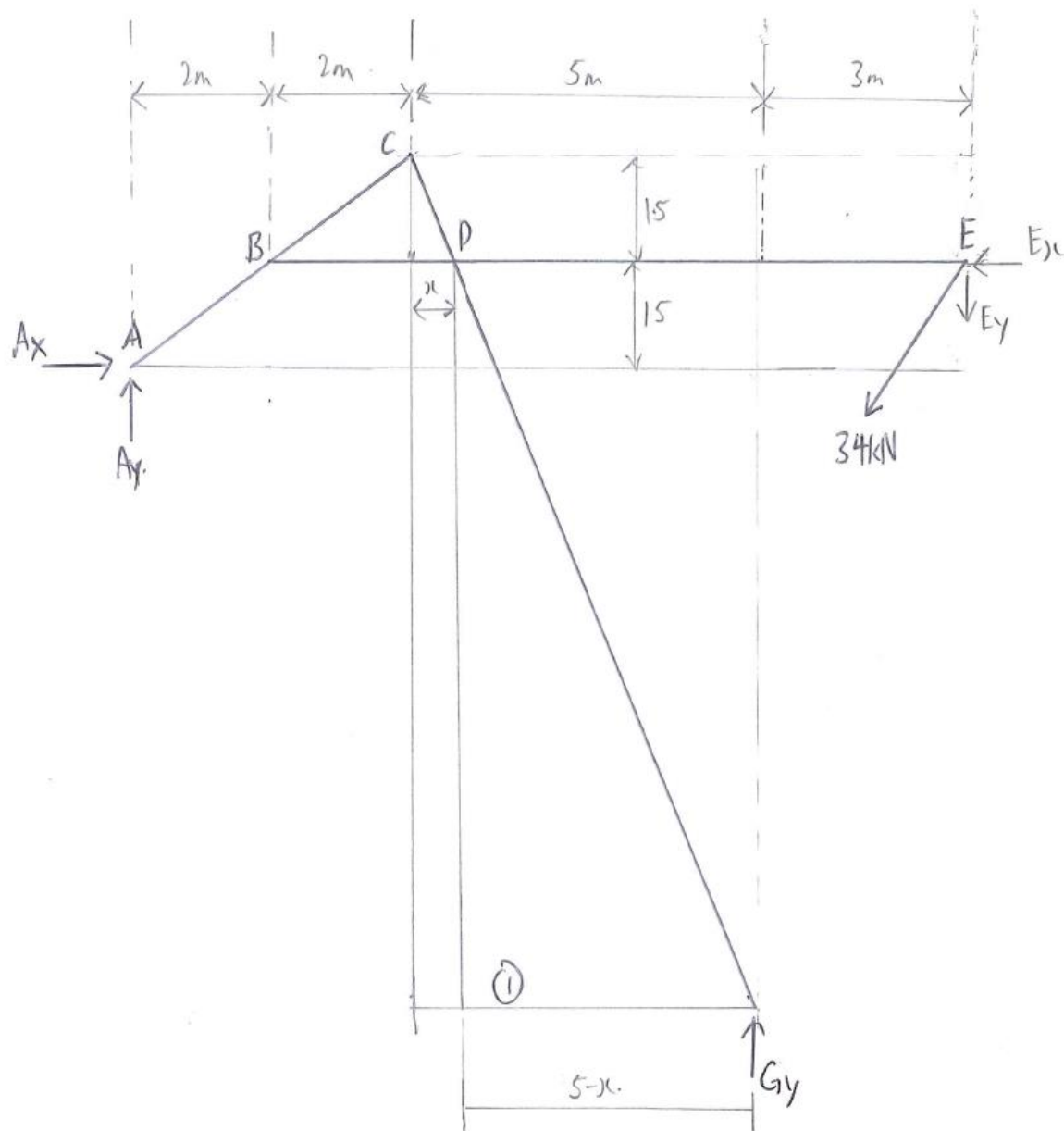
Question 3



Question 3



Question 3



FBD ①:

$$E_y = \frac{15}{\sqrt{8^2 + 15^2}} \cdot 34k$$

$$= 30kN //$$

$$E_x = \frac{8}{\sqrt{8^2 + 15^2}} \cdot 34k$$

$$= 16kN //$$

$$\frac{12}{5} = \frac{10 \cdot 5}{5-x}$$

$$5-x = 4.375$$

$$x = 5 - 4.375$$

$$= 0.625m //$$

Question 3



(3)

FBD (3):

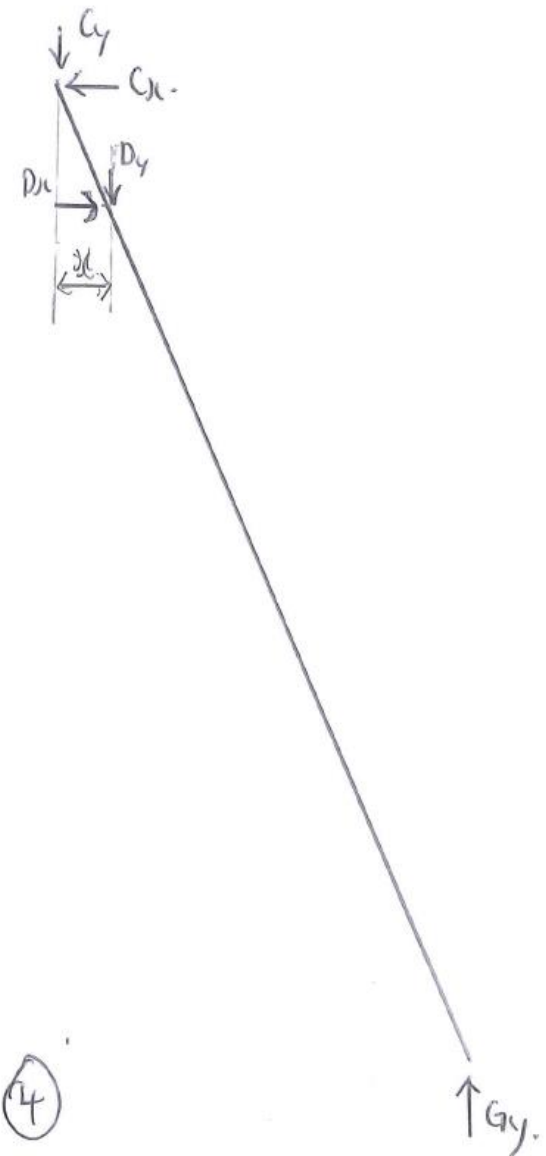
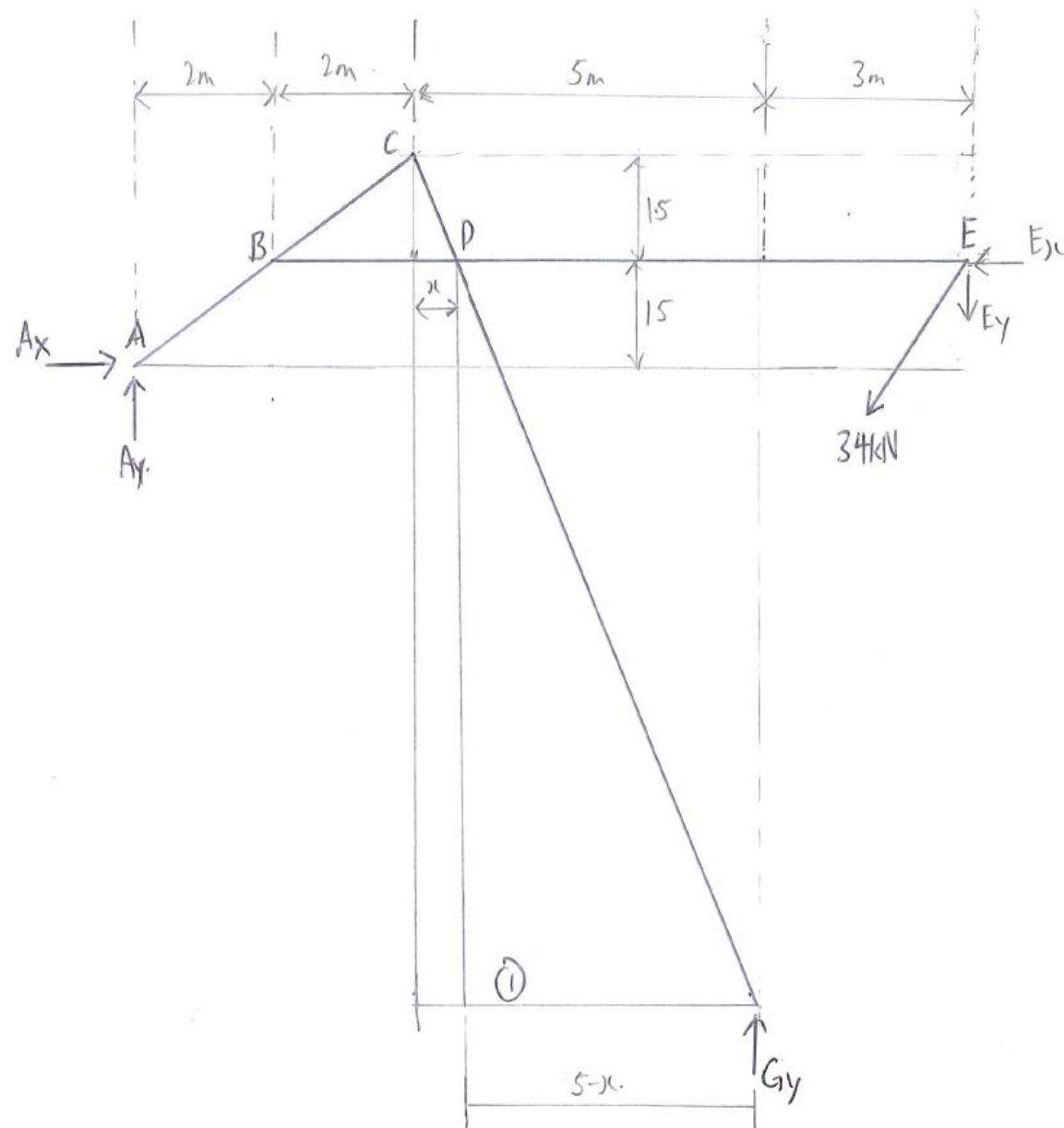
$$\sum M_B = 0$$

$$+\vee : -D_y(2+0.625) + E_y(10) = 0$$

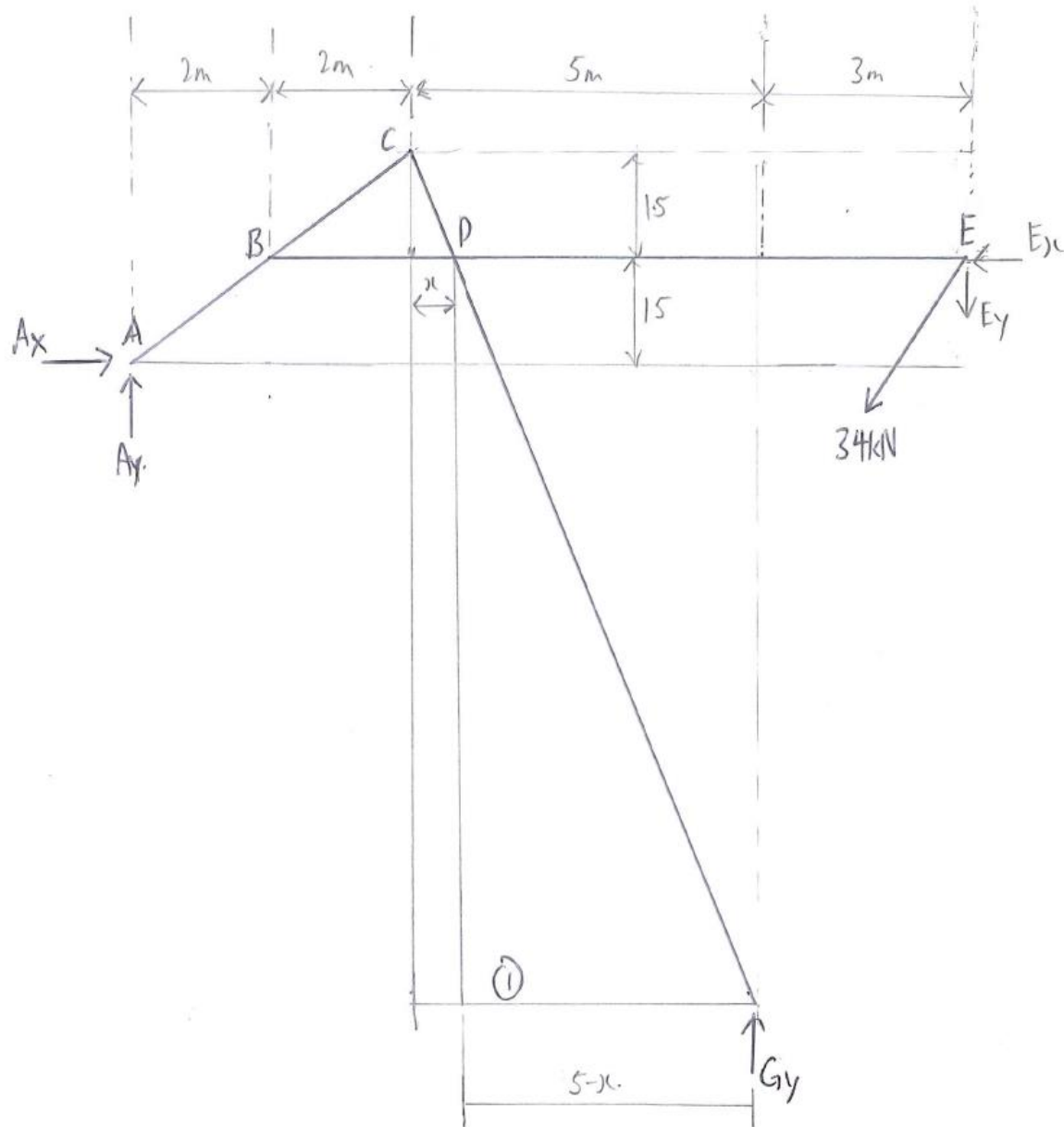
$$2.625 D_y = 300 \text{ k}$$

$$D_y = 114.29 \text{ kN/f}$$

Question 3



Question 3



FBD ①:

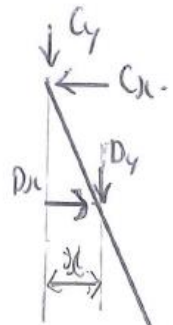
$$\sum M_A = 0$$

$$+\vee \curvearrowleft : 12E_y - 1.5E_x - 9G_y = 0$$

$$9G_y = (30k)(12) - (16k)(1.5)$$

$$G_y = 37.33 \text{ kN} //$$

Question 3



FBD (4):

$$\sum M_c = 0$$

$$+ve \curvearrowleft : (0.625)(D_y) - (1.5)(D_x) - 5(G_y) = 0$$

$$1.5 D_x = (0.625)(114.29k) - (5)(37.33k)$$

$$D_x = -76.81 \text{ kN} //$$

Question 4

4. Determine the force P for the system shown in Figure 2.7 to be in static equilibrium. What are the horizontal and vertical components of the pin reactions at A and B?

Ans: $P = 5.21 \text{ kN}$; $A_x = 2 \text{ kN}$; $A_y = 6.66 \text{ kN}$; $B_x = 5 \text{ kN}$; $B_y = 1.45 \text{ kN}$

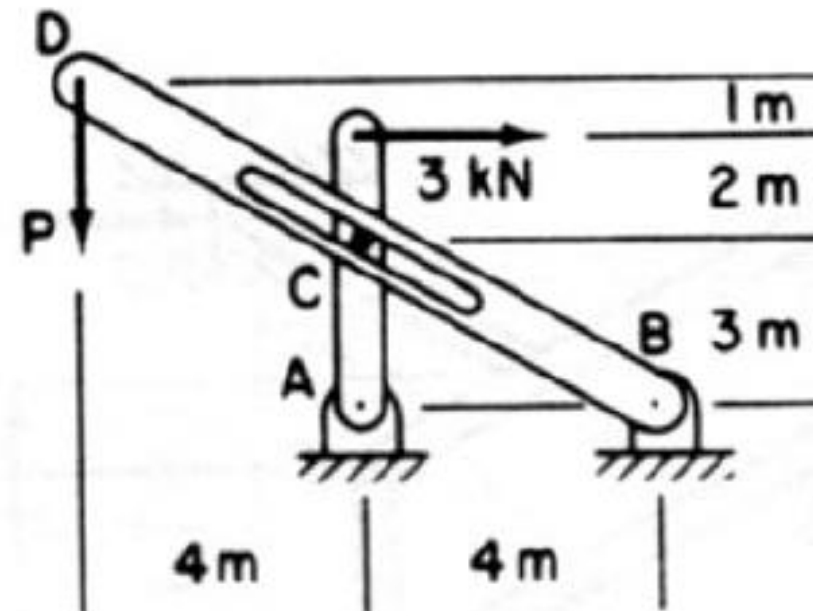


Figure 2.7

Question 4

4. Determine the force P for the system shown in Figure 2.7 to be in static equilibrium. What are the horizontal and vertical components of the pin reactions at A and B?

Ans: $P = 5.21 \text{ kN}$; $A_x = 2 \text{ kN}$; $A_y = 6.66 \text{ kN}$; $B_x = 5 \text{ kN}$; $B_y = 1.45 \text{ kN}$

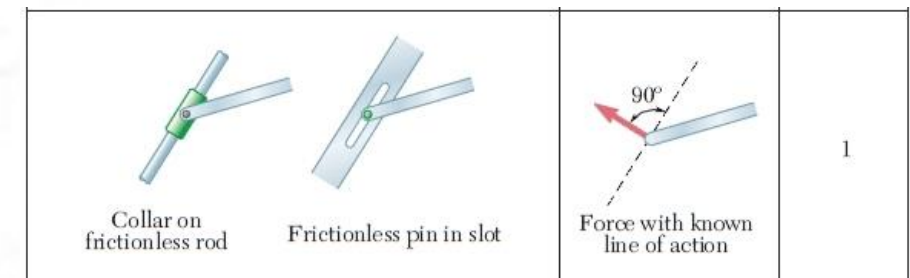
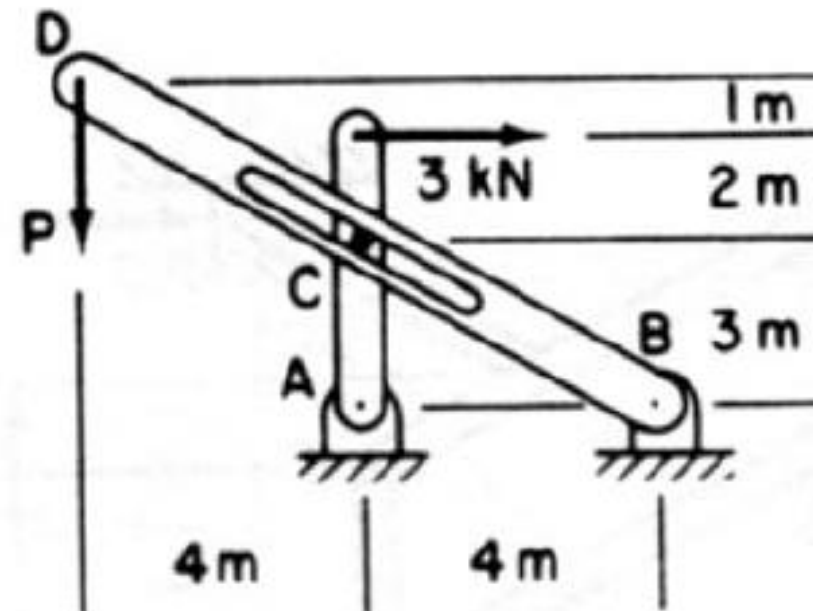
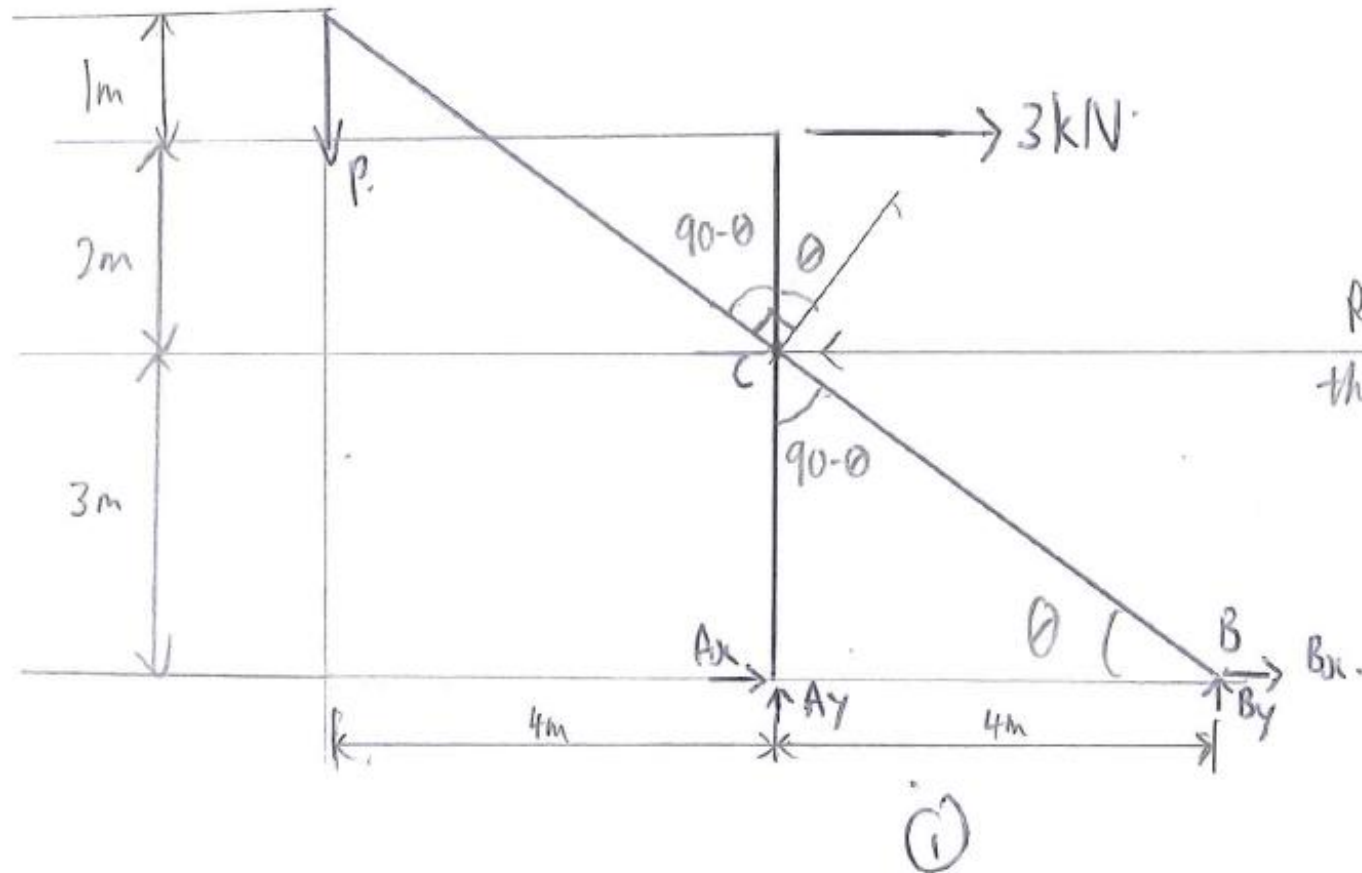


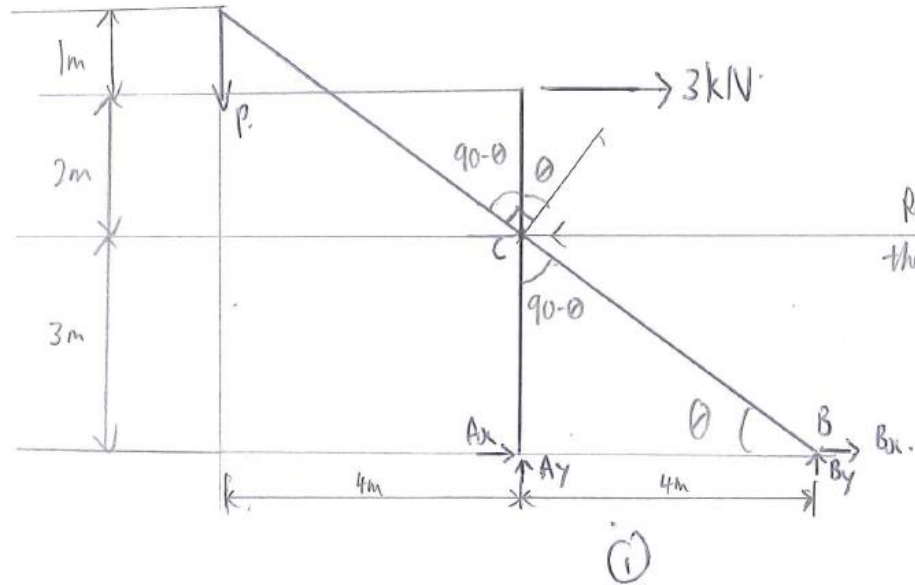
Figure 2.7

Question 4

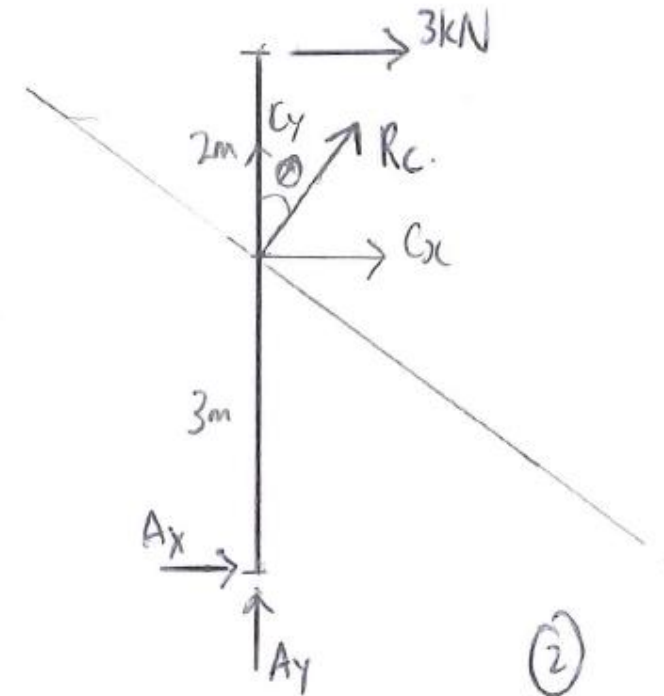


Remember for pin-slot,
the reaction force is perpendicular to the slot.

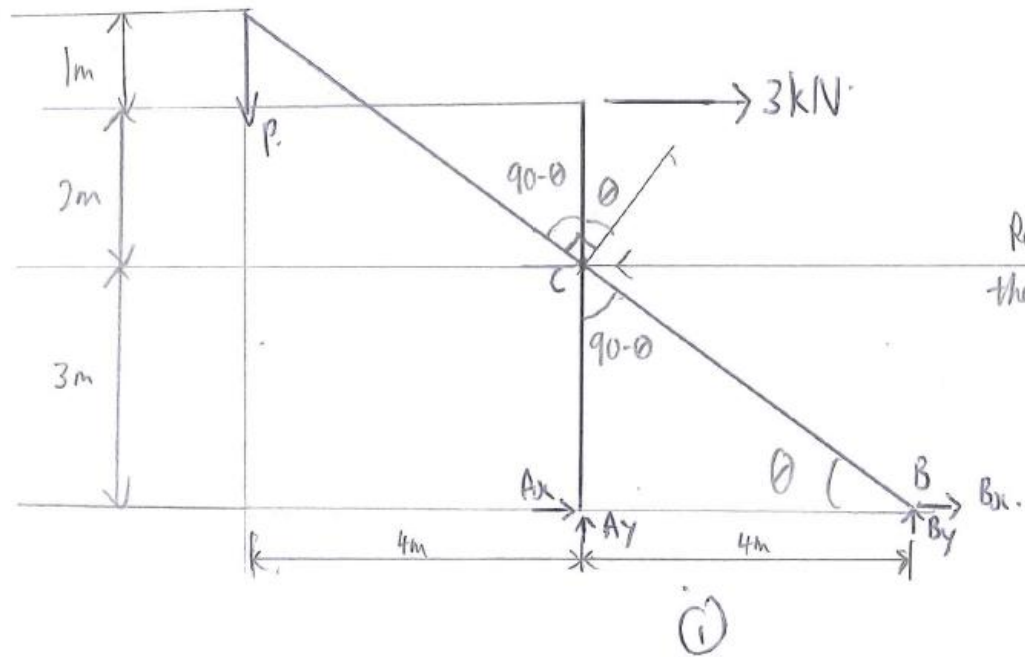
Question 4



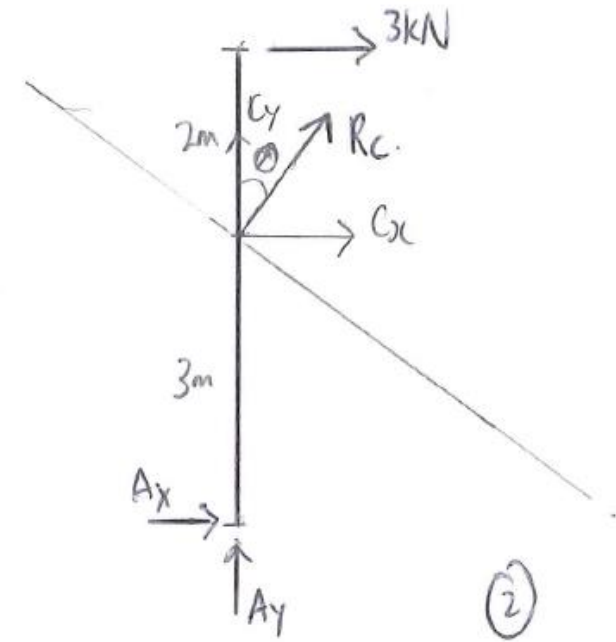
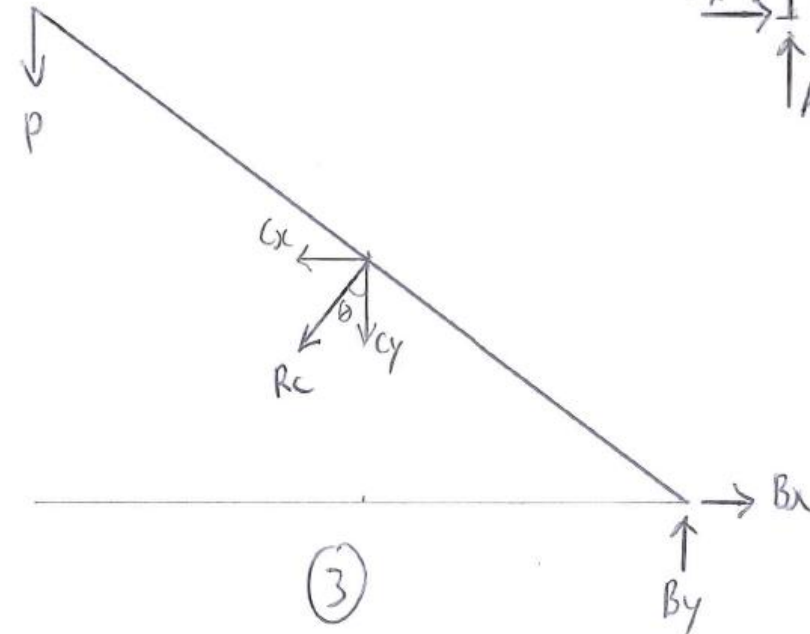
Remember for pin-slot,
the reaction force is h to the slot.



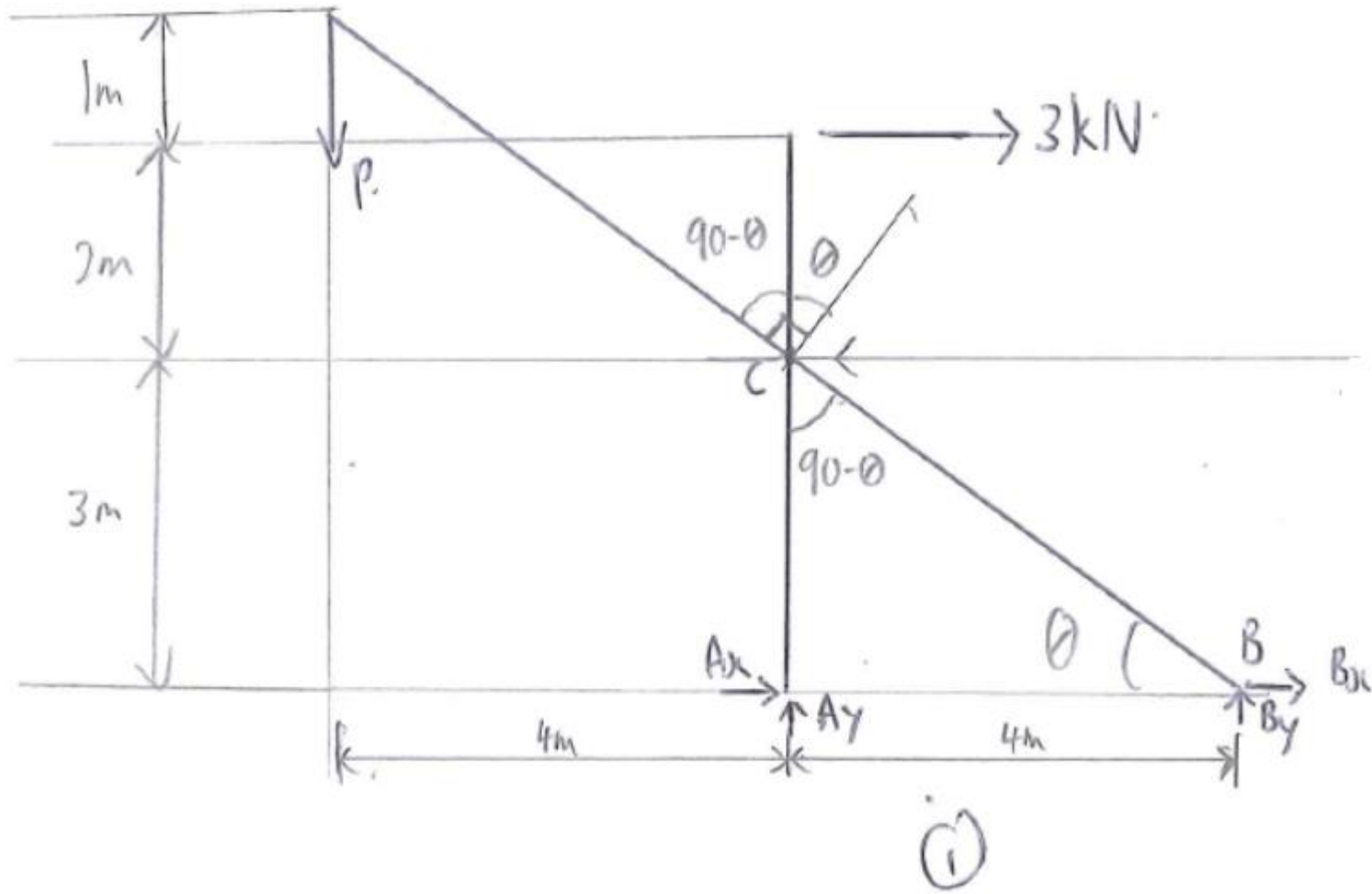
Question 4



Remember for pin-slot,
the reaction force is h to the slot.



Question 4



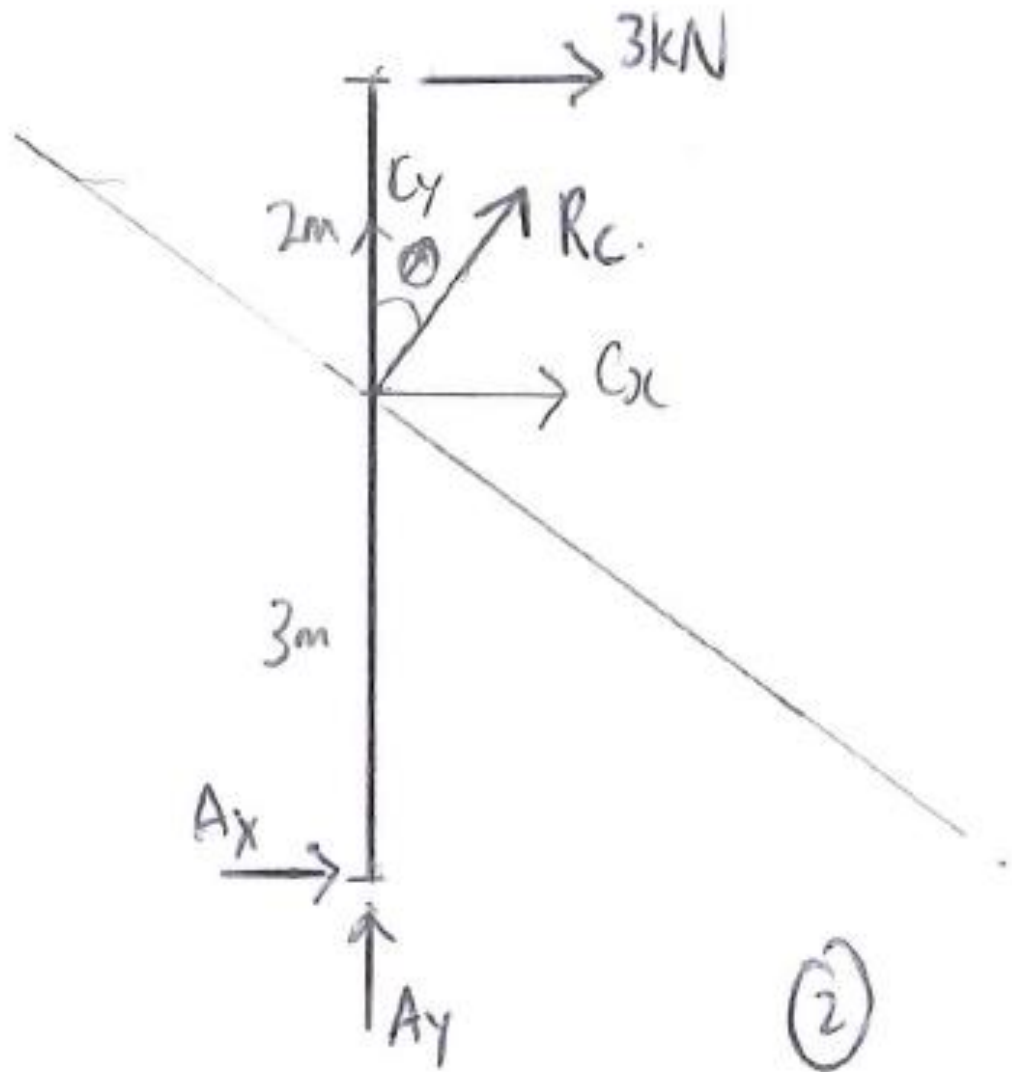
FBD ①:

$$\tan \theta = \frac{3}{4}$$

$$\theta = \tan^{-1} 0.75$$

$$= 36.87^\circ$$

Question 4



FBD (2):

$$\sum M_A = 0$$

$$(3)(C_x) + (3+2)(3k) = 0$$

$$C_x = -5kN_{//}$$

$$\Rightarrow \tan 36.87^\circ = C_x / C_y$$

$$C_y = \frac{C_x}{\tan 36.87^\circ}$$

$$= -6.66 kN_{//}$$

$$\sum F_y = 0$$

$$A_y + C_y = 0$$

$$A_y = -C_y$$

$$= 6.66 kN_{//}$$

$$\sum F_x = 0$$

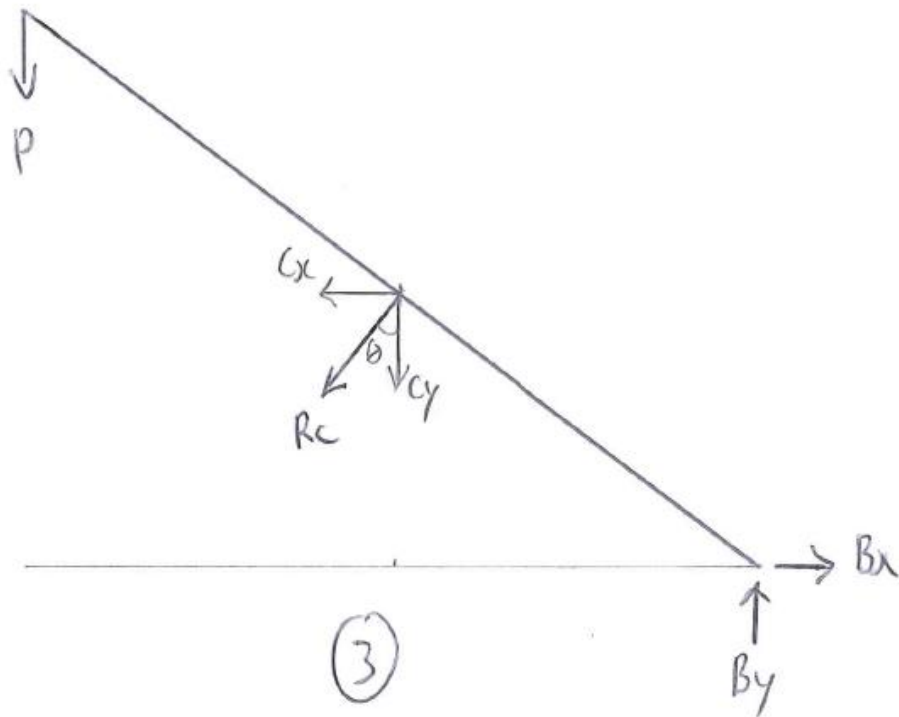
$$A_x + C_x + 3k = 0$$

$$A_x = -C_x - 3k$$

$$= 5k - 3k$$

$$= 2kN_{//}$$

Question 4



FBD (3).

$$\sum M_B = 0$$

$$\text{+ve} \quad -(P)(8) - (C_x)(3) - (C_y)(4) = 0$$

$$8P = -(3)(-5\text{k}) - (4)(-6.66)$$

$$P = 5.21\text{kN} //$$

$$\sum F_x = 0$$

$$B_x - C_x = 0$$

$$B_x = C_x$$

$$= -5\text{kN} //$$

$$\sum F_y = 0$$

$$B_y - C_y - P = 0$$

$$B_y = P + C_y$$

$$= 5.21\text{k} - 6.66\text{k}$$

$$= -1.45\text{kN} //$$

Question 5

5. Three bars are connected with smooth pins form the A-frame shown in Figure 2.8. The weights of the bars can be neglected. Determine the forces exerted by the pins at B and D on member BD and the force exerted by the pin at C on member CDE.

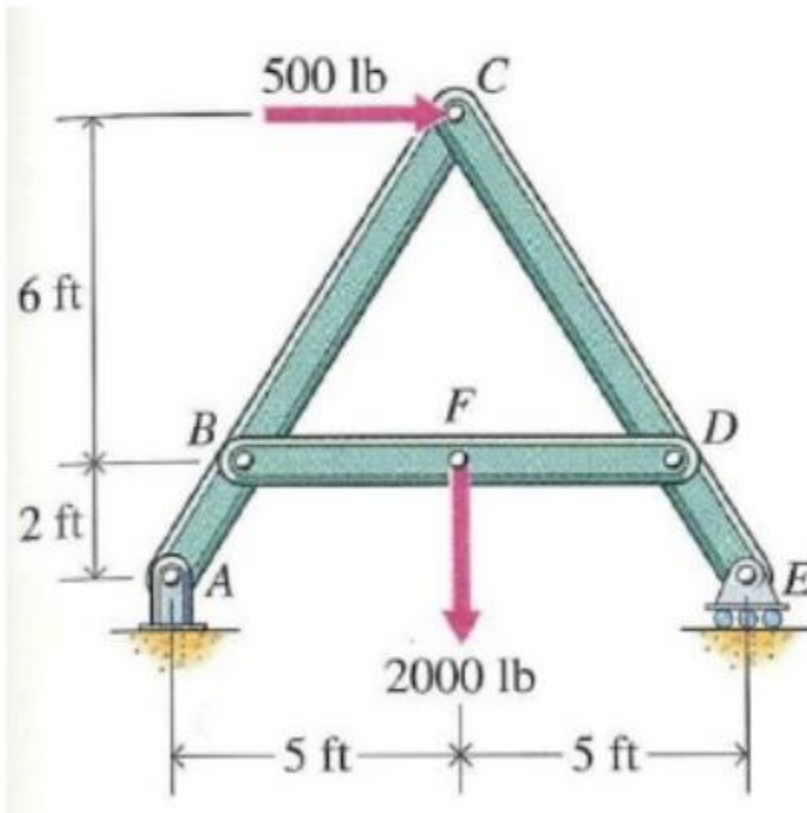
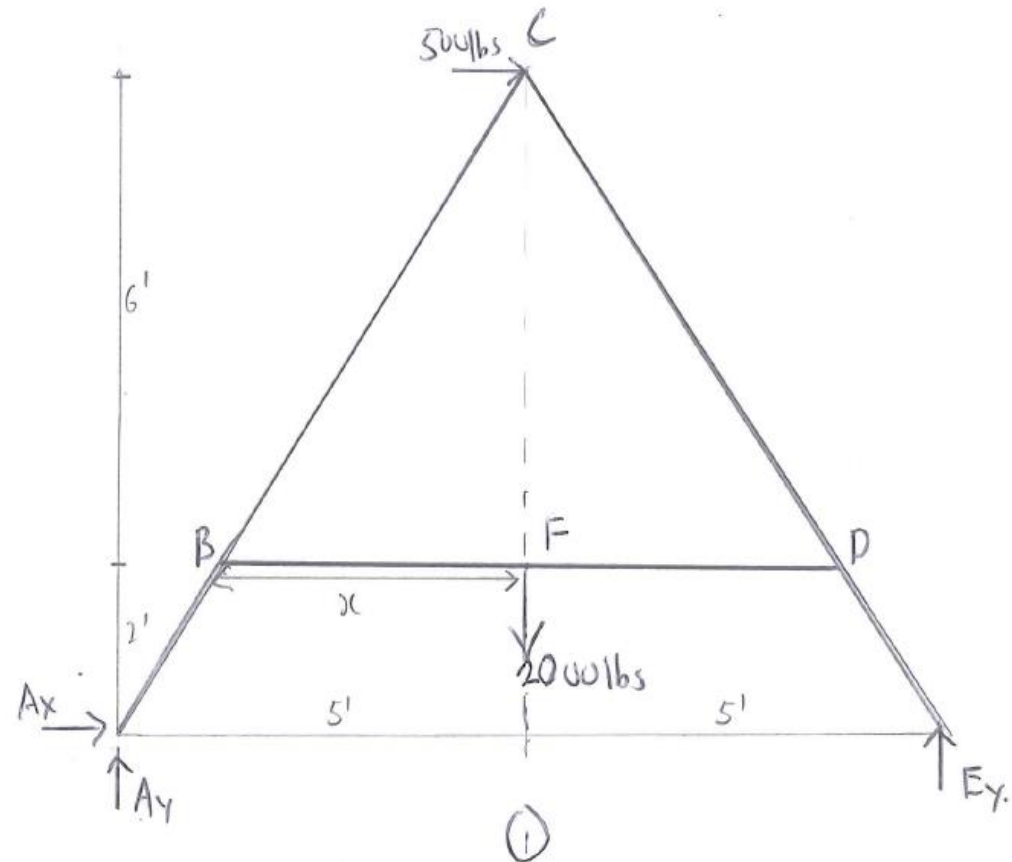
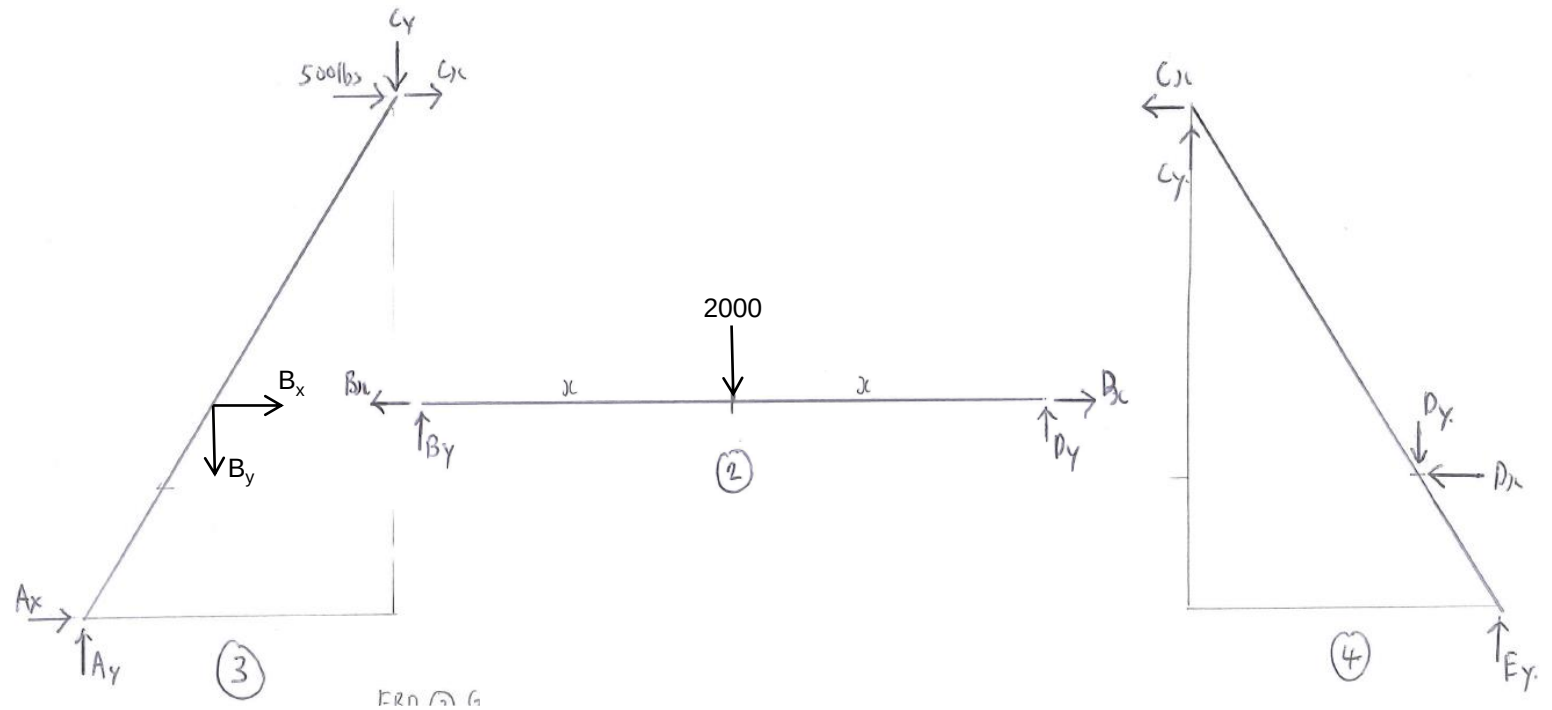
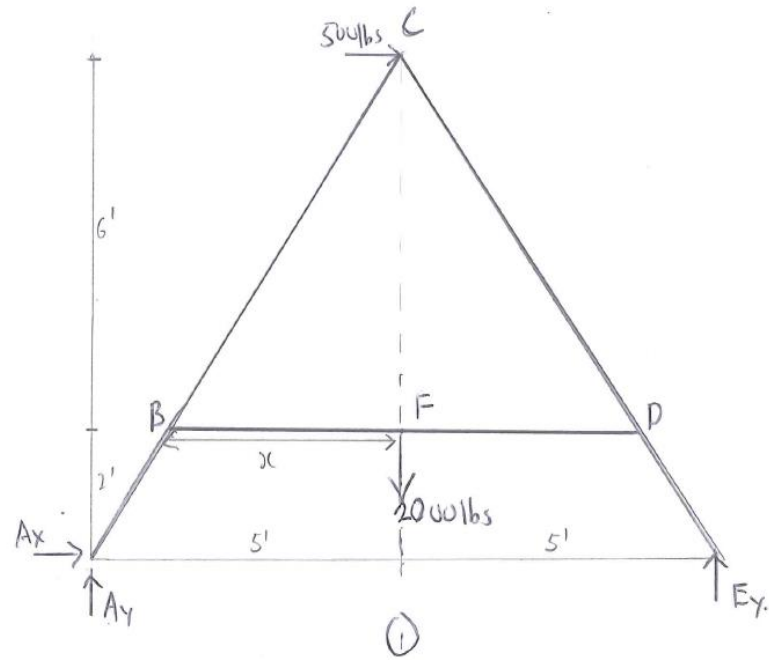


Figure 2.8

Question 5

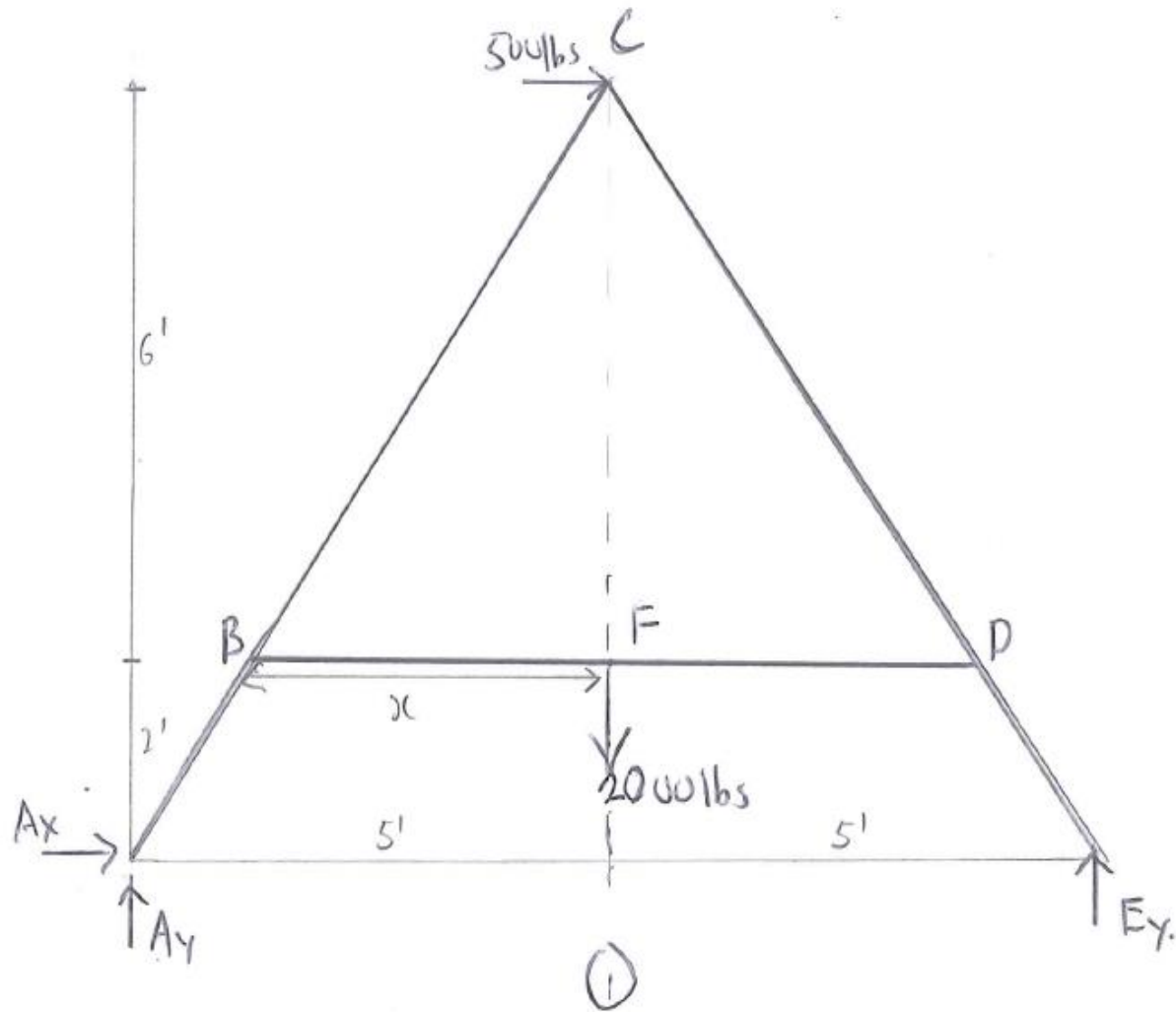


Question 5



FBD ①, ③ & ④ must
join together to become
FBD ① so the 500lbs only appear once.

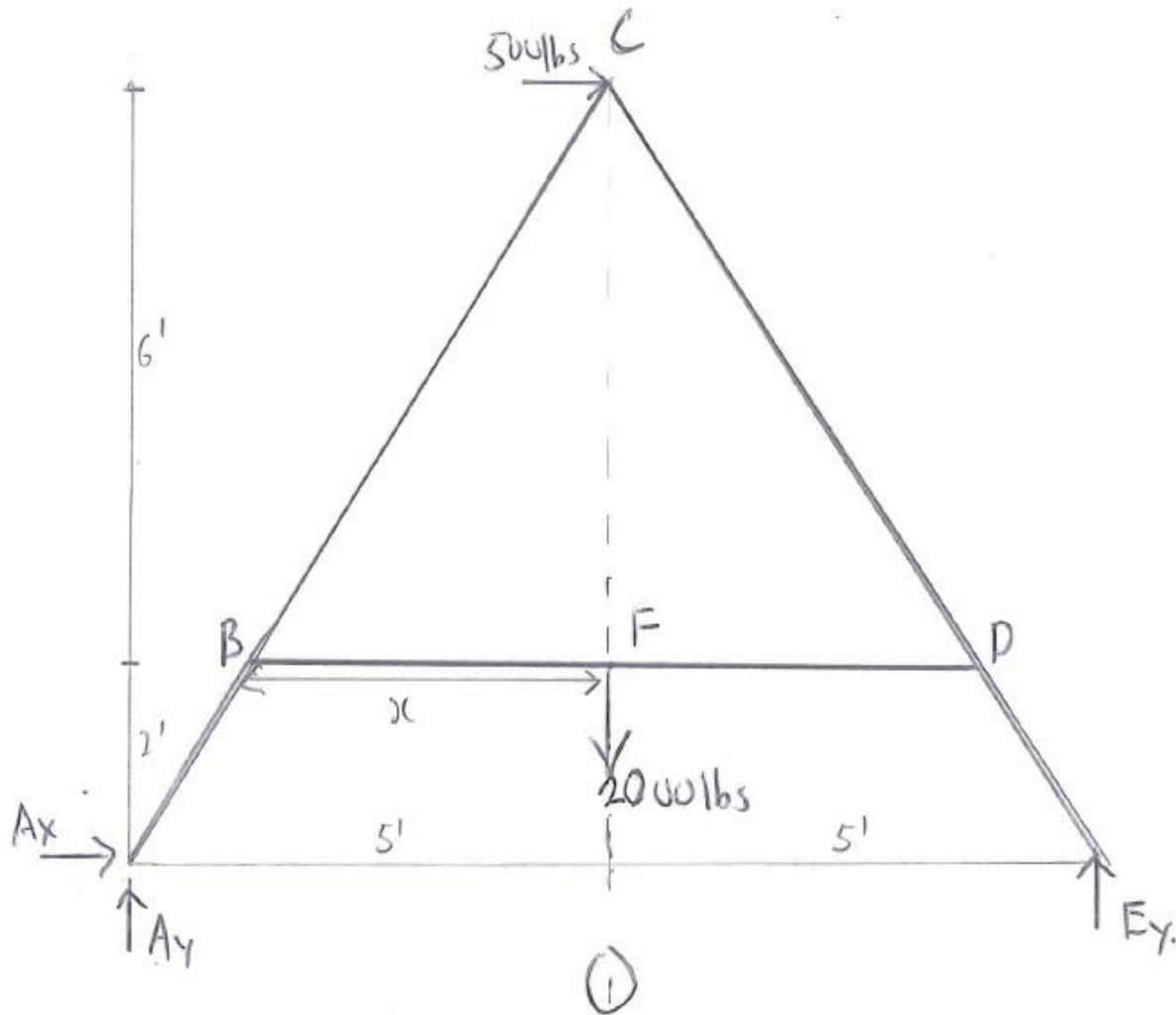
Question 5



$$\frac{x}{6} = \frac{5}{8}$$

$$x = 3.75'$$

Question 5



FBP ①:

$$\sum F_{x_i} = 0$$

$$A_{x_i} + 500 = 0$$

$$A_{x_i} = -500 \text{ lbs} //$$

$$\sum M_A = 0$$

$$\text{FBD } \textcircled{2}: (500)(8) + 2000(5) - (E_y)(10) = 0$$

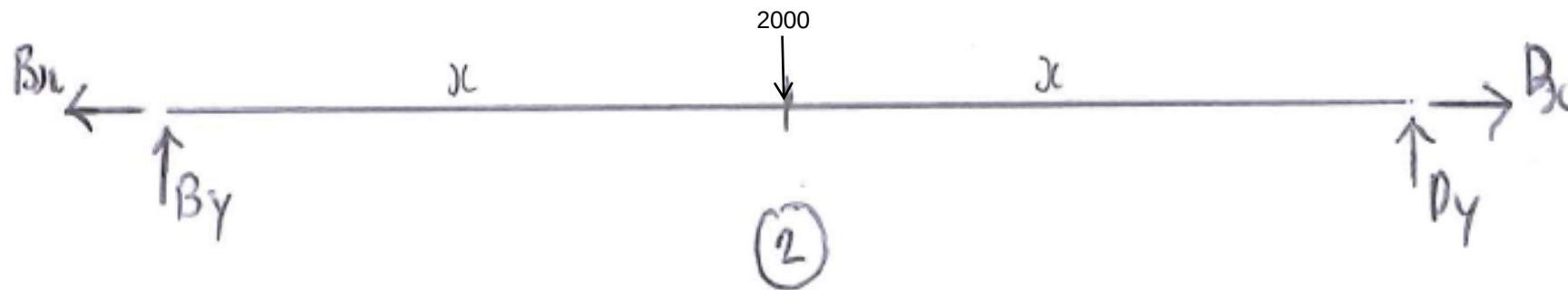
$$E_y = 1400 \text{ lbs} //$$

$$\sum F_y = 0$$

$$1400 + A_y - 2000 = 0$$

$$A_y = 600 \text{ lbs} //$$

Question 5



FBD (2):

$$\sum M_B = 0$$

$$(2000)(3.75) - (3.75 + 3.75)(D_y) = 0$$

$$D_y = 1000 \text{ lbs.} //$$

$$\sum F_y = 0$$

$$B_y + D_y - 2000 = 0$$

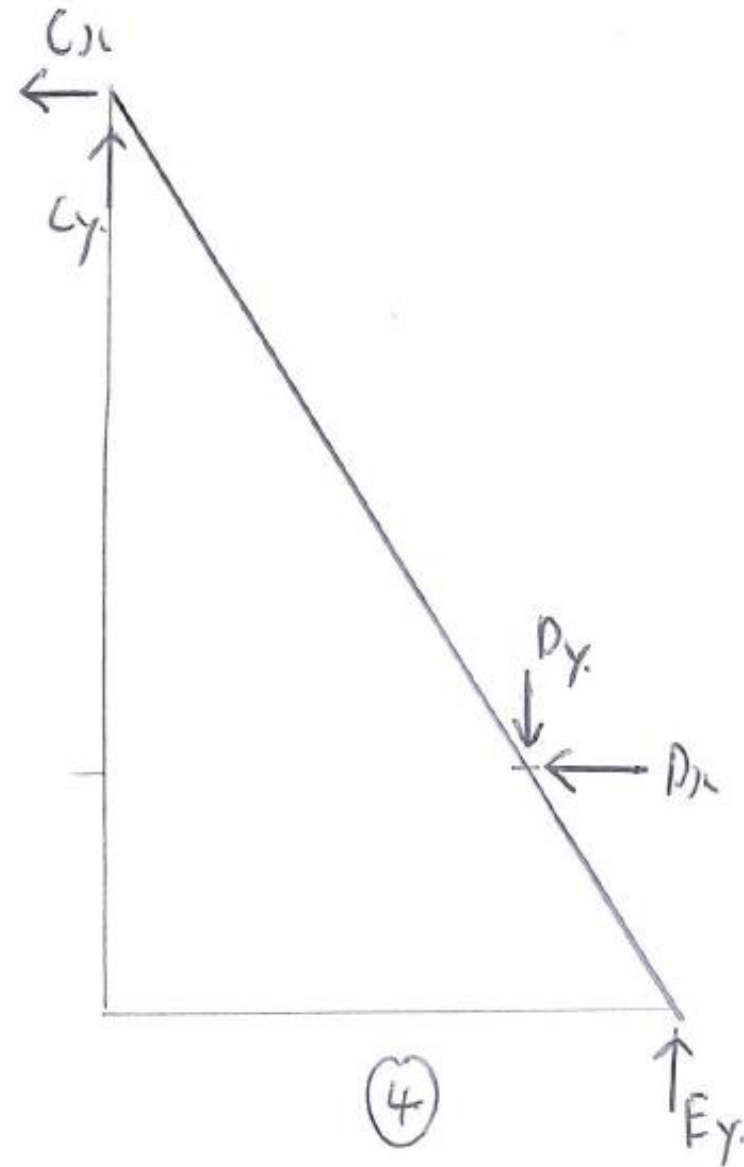
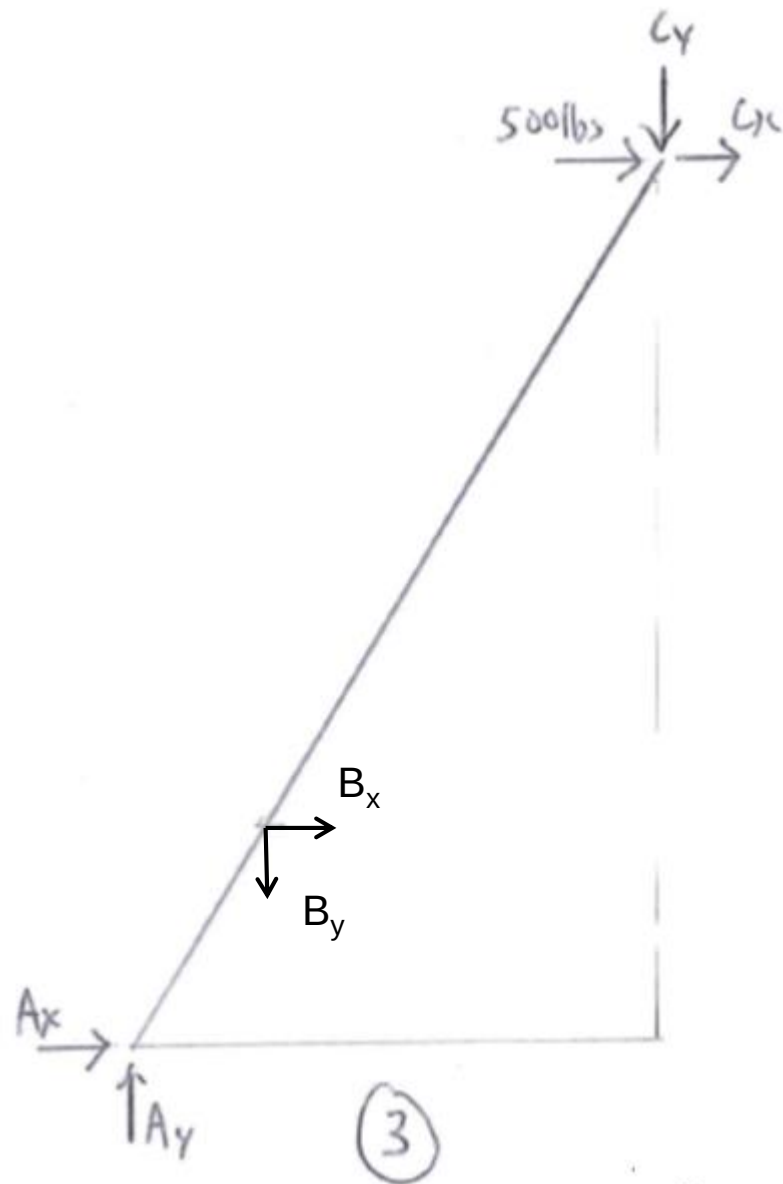
$$\begin{aligned} B_y &= 2000 - 1000 \\ &= 1000 \text{ lbs.} // \end{aligned}$$

$$\sum F_x = 0$$

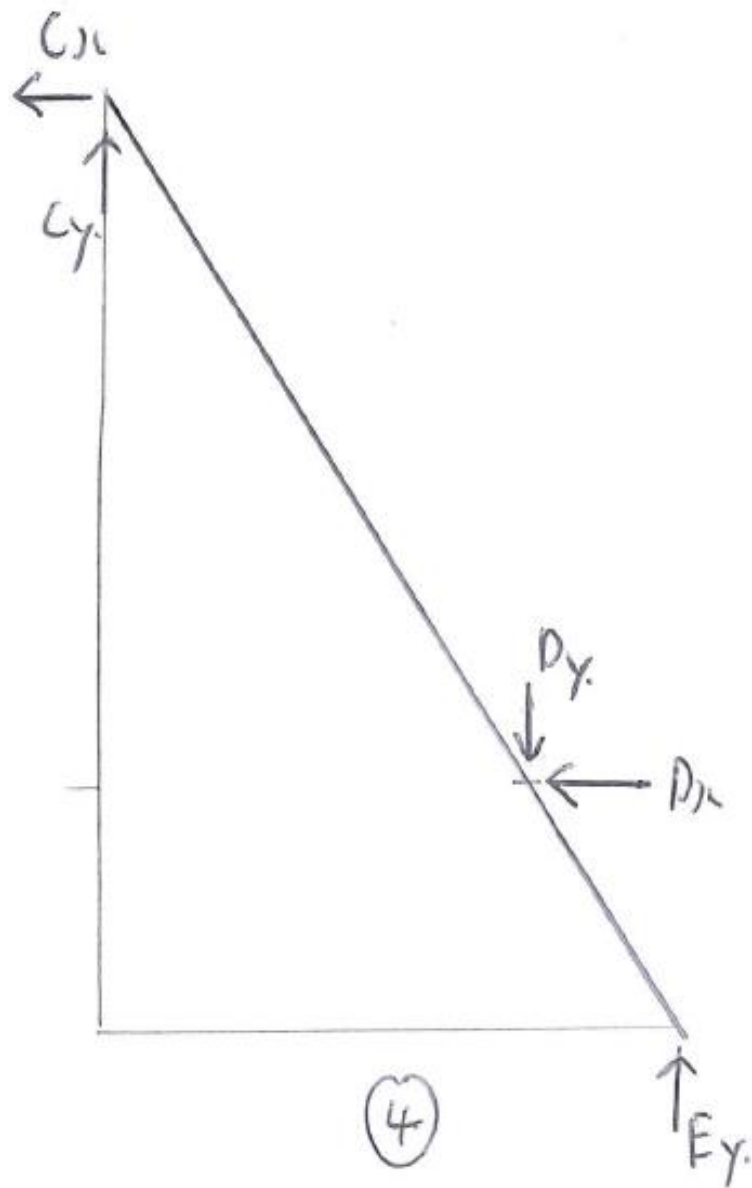
$$B_x - D_x = 0$$

$$B_x = D_x //$$

Question 5



Tutorial 2 Question 5



FBD (4):

$$\sum M_C = 0$$

$$+\vee \downarrow (3.75)(D_y) + (6)(D_x) - (5)(E_y) = 0$$

$$6D_x = (5)(E_y) - (3.75)(D_y)$$

$$D_x = 541.67 \text{ lbs//}$$

$$\Rightarrow B_x = 541.67 \text{ lbs//}$$

$$\sum F_y = 0$$

$$C_y + E_y - D_y = 0$$

$$C_y = D_y - E_y$$

$$= 1000 - 1400$$

$$= -400 \text{ lbs//}$$

$$\sum F_x = 0$$

$$C_x + D_x = 0$$

$$C_x = -D_x$$

$$= -541.67 \text{ lbs//}$$

Question 6

6. A section of a landing gear support system is simplified into the system shown in Figure 2.9. The system is pivoted at point A, B and D. A L-shaped bracket, BD (two-force member), was designed and installed on member ABC due to space constraints. The weight of the system can be neglected and the diagram is not drawn to scale.

- Determine the reaction(s) at D. ($D_x=1313.87 \text{ kN}$; $D_y = 525.51 \text{ kN}$)
- Hence, determine if the square bracket BD is under compression or tension and explain the reason. (**Tension**)
- Determine the reaction(s) at A. ($A_x=1313.87 \text{ kN}$; $A_y = 324.48 \text{ kN}$)

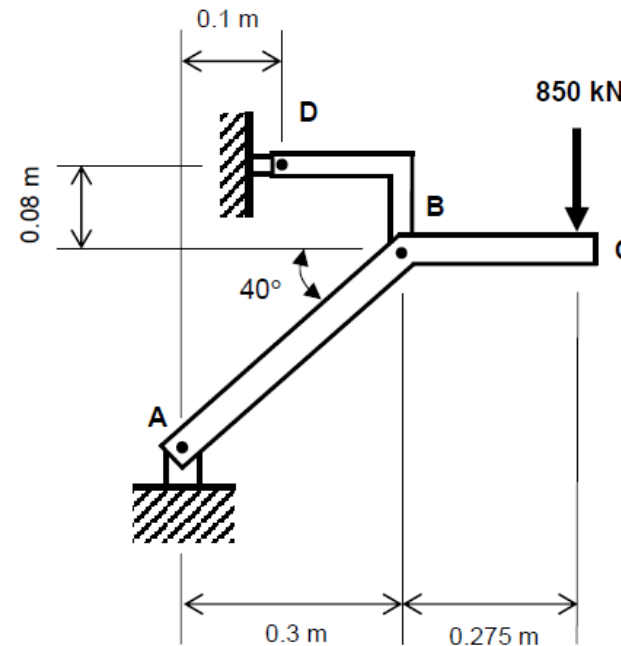
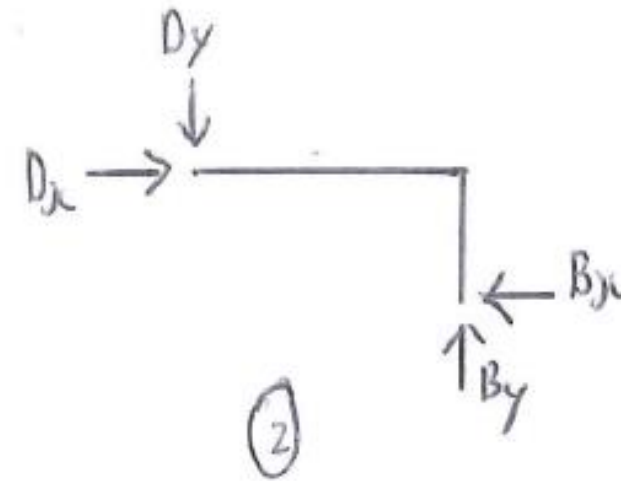
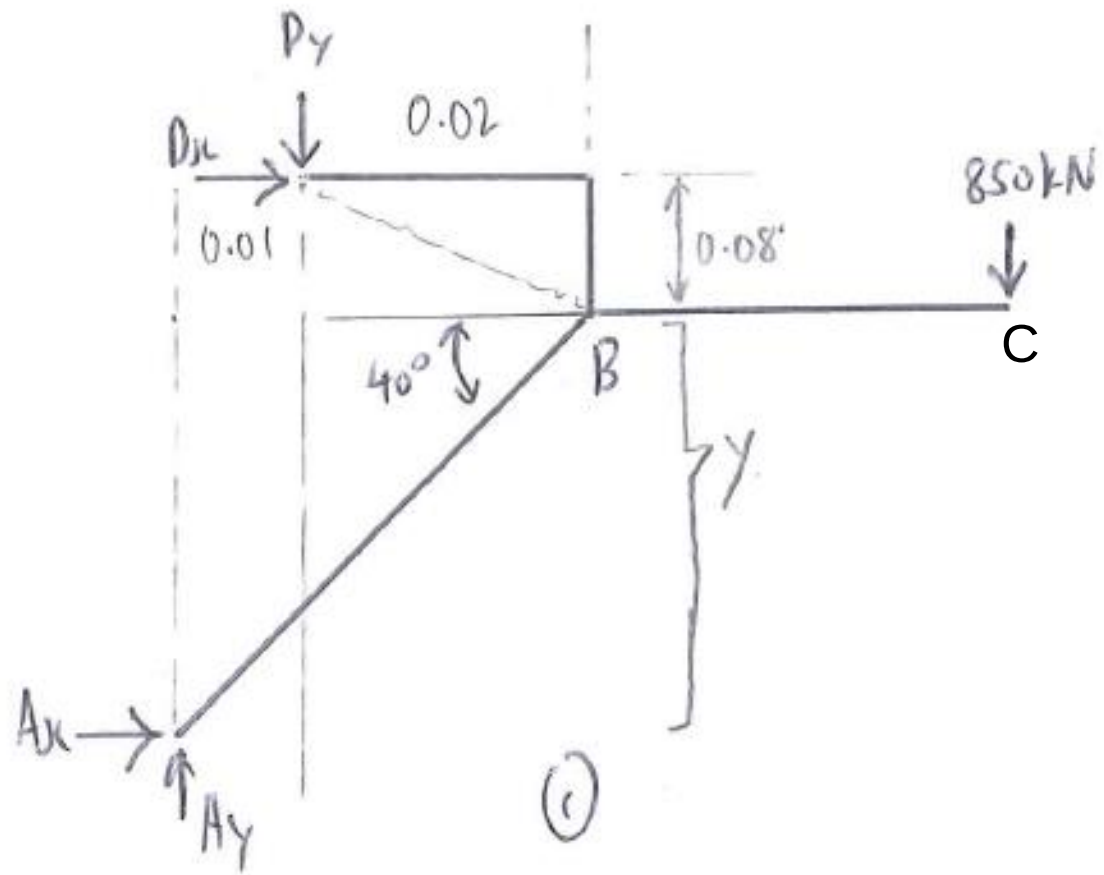
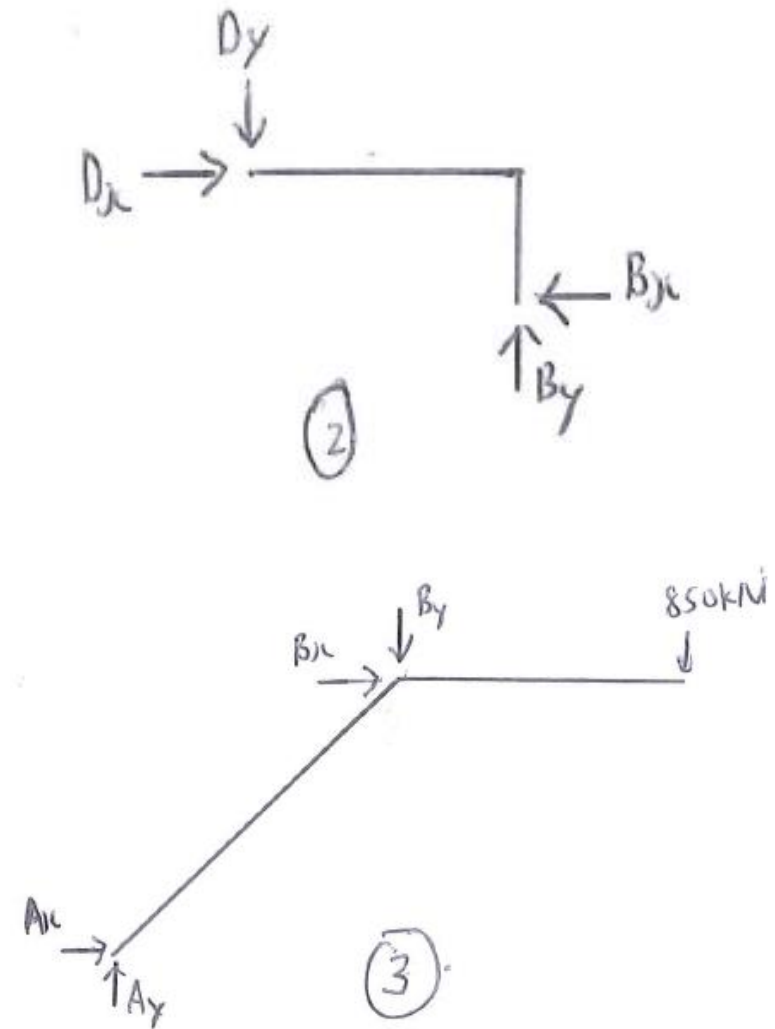
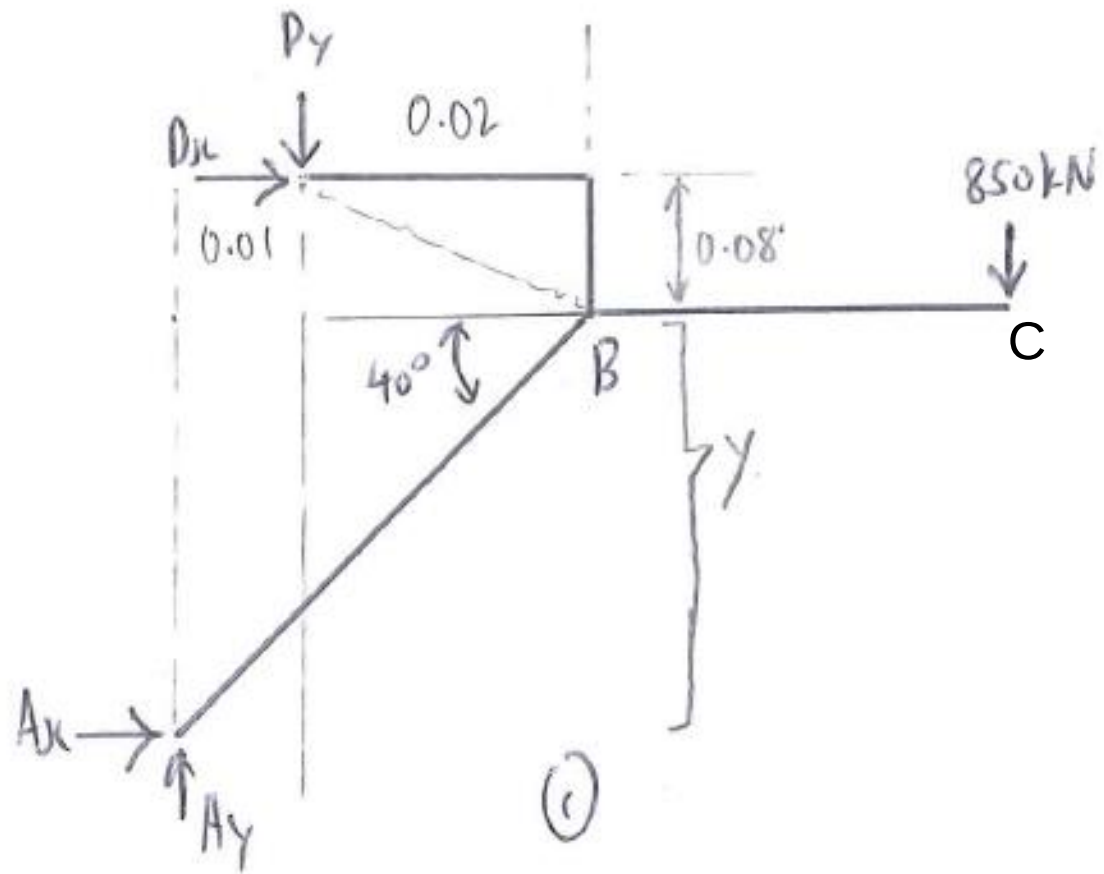


Figure 2.9

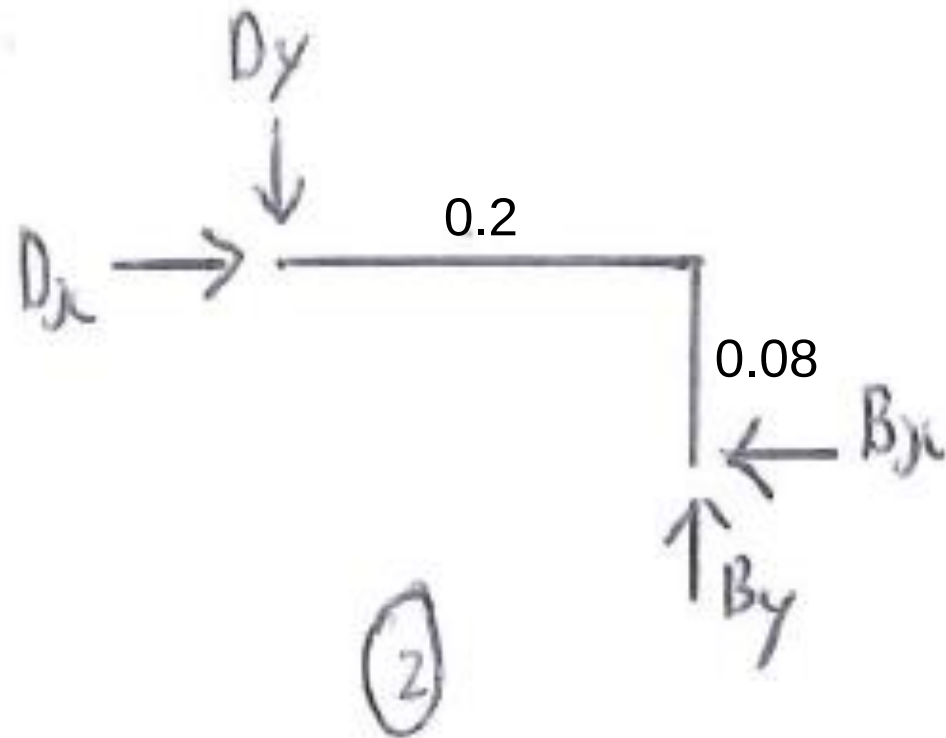
Question 6



Question 6



Question 6



FBD (2):

$$\frac{D_y}{D_x} = \frac{0.08}{0.2}$$

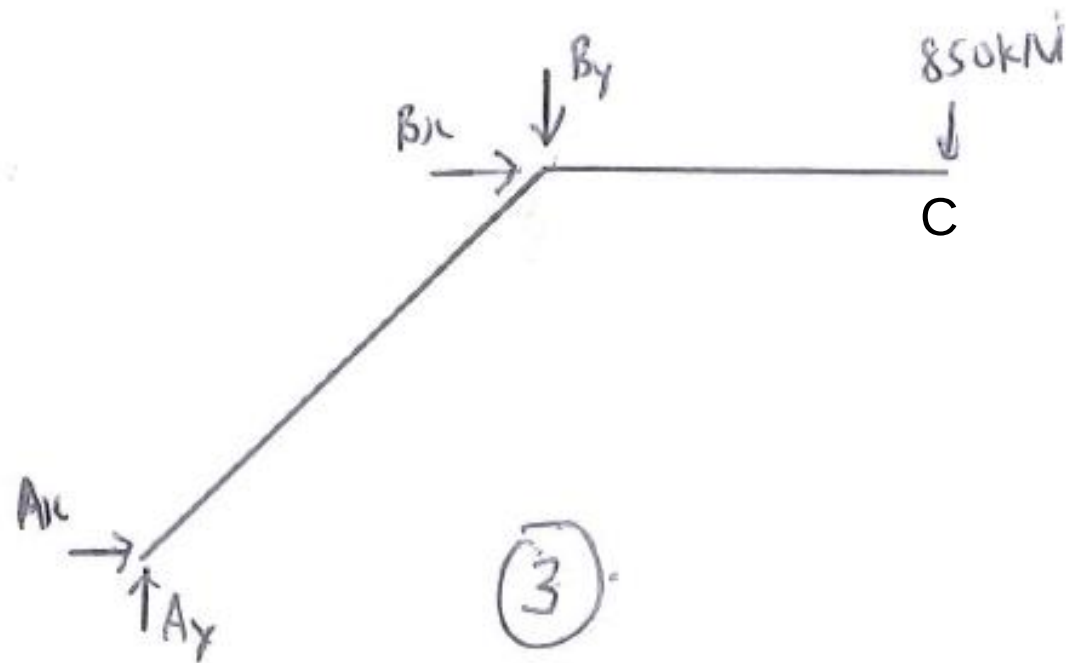
$$D_y = 0.4 D_x$$

$$\text{or } D_x = 2.5 D_y$$

Also $D_x = B_x$
 $D_y = B_y$

$$\left(\begin{array}{l} D_x = B_x \\ D_y = B_y \\ \Rightarrow B_y = 0.4 B_x \end{array} \right)$$

Question 6



FBD ③:

$$\sum M_A = 0$$

$$\begin{aligned} \text{+ve} \curvearrowleft: (850\text{k})(0.3 + 0.275) + (B_y)(0.3) + (B_x)(0.2517) &= 0 \\ (850\text{k})(0.575) + (0.4B_x)(0.3) + 0.2517B_x &= 0 \\ (0.3717B_x) &= -488.75\text{kN} \end{aligned}$$

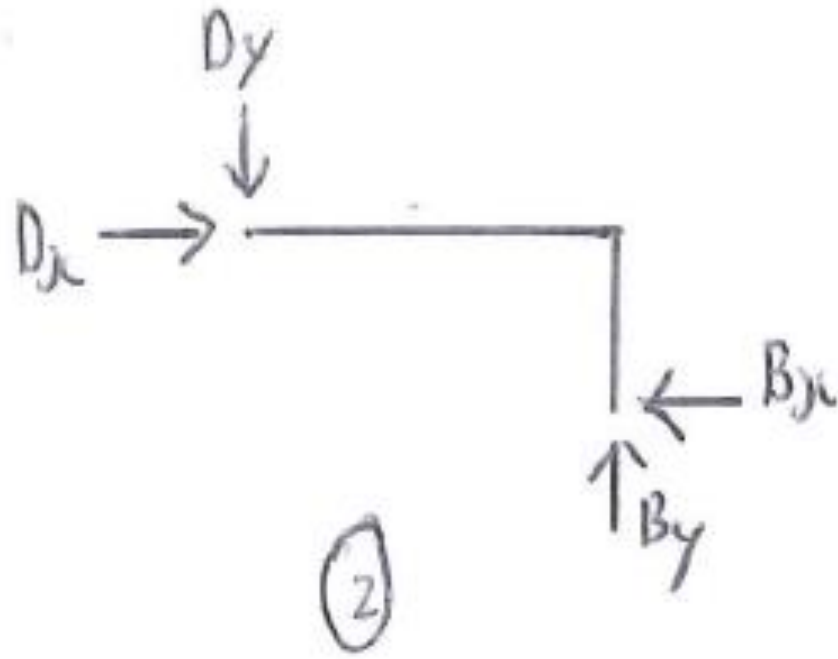
$$B_x = -1314.9\text{ kN}_{//}$$

$$\Rightarrow D_x = -1314.9\text{ kN}_{//}$$

$$B_y = -525.96\text{ kN}_{//}$$

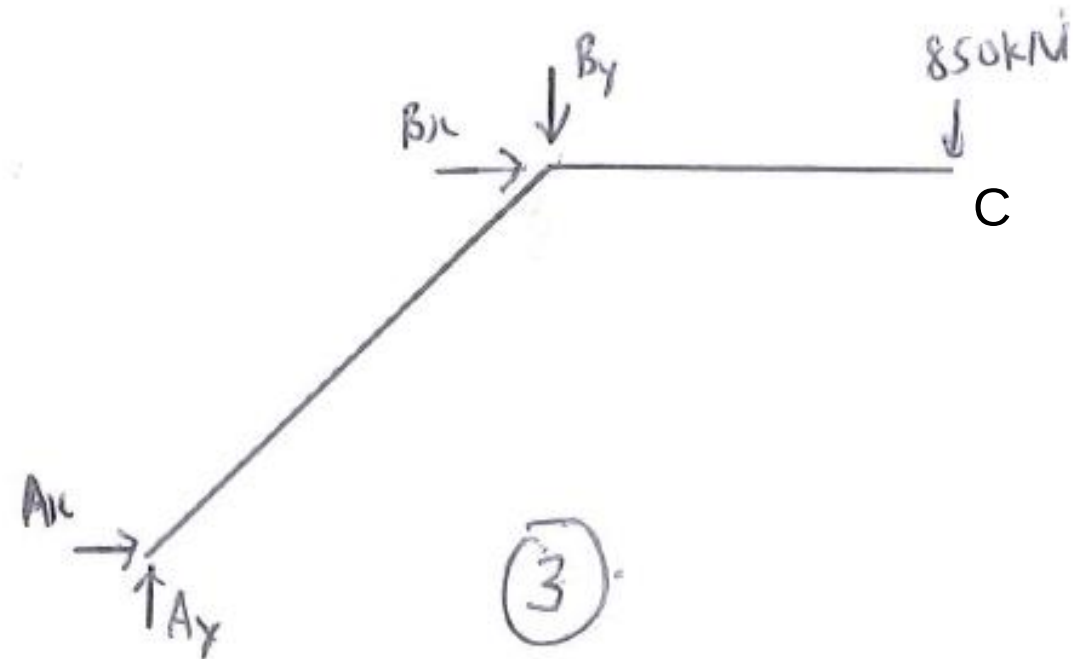
$$D_y = -525.96\text{ kN}_{//}$$

Question 6



$$B_x = -1314.9 \text{ kN}_{//}$$

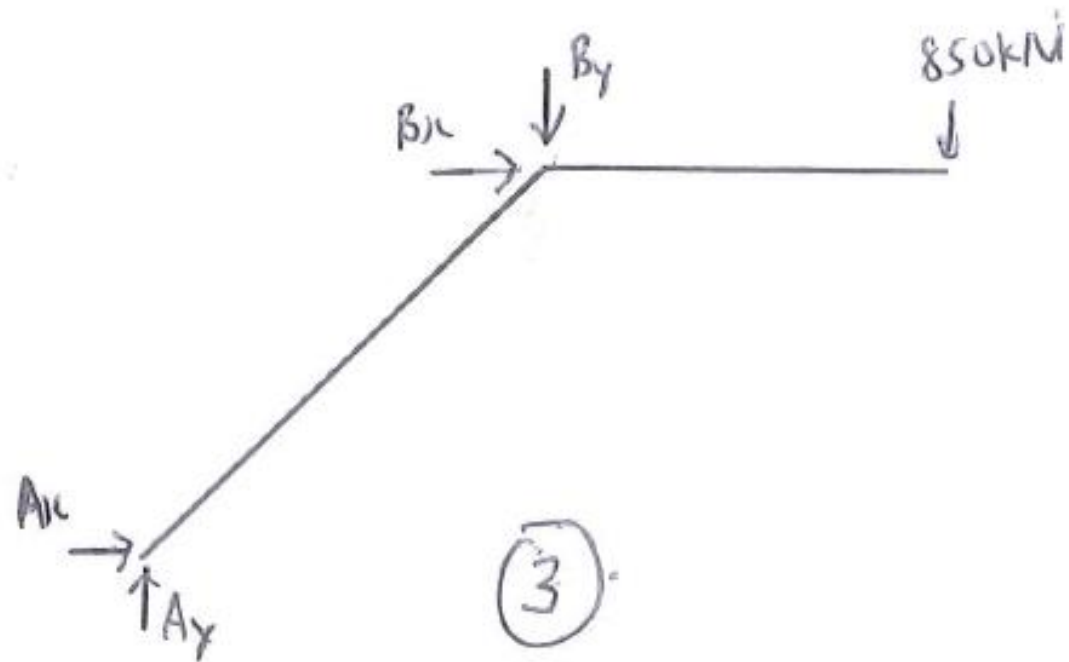
$$B_y = -525.96 \text{ kN}_{//}$$



$$\Rightarrow D_x = -1314.9 \text{ kN}_{//}$$

$$D_y = -525.96 \text{ kN}_{//}$$

Question 6



FBD (3)

$$\sum F_y = 0$$

$$A_y - B_y - 850 \text{ k} = 0$$

$$A_y = 850 \text{ k} - 525.96 \text{ k}$$

$$= 324.04 \text{ kN} //$$

$$\sum F_x = 0$$

$$A_x + B_x = 0$$

$$A_x = -B_x$$

$$= 1314.9 \text{ kN} //$$

The End